

Quantifying the Influence of Fundus Image Quality on Automated Diabetic Retinopathy Severity Detection

Omar Khalid Alghamdi

Module/Code: KV6013 Individual Computing Project

Programme: BSc (Hons) Artificial intelligence

Word count: 11966 (excluding title page, contents, references and appendix)

Abstract—Diabetic retinopathy (DR) is a major complication of diabetes and an important cause of preventable vision impairment worldwide. Artificial intelligence (AI)-based retinal image analysis has shown strong potential for automated DR grading, but variations in image quality, particularly blur in low-cost or smartphone-based fundus imaging, may affect reliability. The impact of blur on DR severity grading and on architectural robustness was investigated. A unified EyePACS, APTOS and Messidor fundus collection distributed through Kaggle [1] was used, labelled on the five-grade international severity scale of no DR, mild, moderate, severe and proliferative DR. A class-capped subset of 7,006 training, 1,574 validation and 1,581 test images kept training within the computational budget. Four architectures were evaluated: a custom CNN baseline, ResNet50, DenseNet201 and EfficientNetB0. All were trained on clean images only and tested on the same held-out images re-rendered at five Gaussian blur levels, from $\sigma = 0$ (unmodified) to $\sigma = 4$ (80% blur), with degradation verified by variance of Laplacian and peak signal-to-noise ratio. Performance was assessed using macro F1, macro ROC-AUC, quadratic weighted kappa (QWK), per-class sensitivity, ordinal error analysis and Grad-CAM visualisation. Blur degraded performance for every model, with pretrained architectures substantially stronger than the CNN baseline. Averaged across the three pretrained models, macro F1 fell by 67.5% and QWK by 88.1% between $\sigma = 0$ and 4, with the steepest losses between $\sigma = 1$ and 3. ResNet50 achieved the highest clean macro F1 of 0.4951 and EfficientNetB0 the highest QWK of 0.6881, but the ordering reversed under degradation: of the three, EfficientNetB0 retained the largest share of its clean performance, ahead of DenseNet201 and then ResNet50. Blur also pushed most models towards overgrading: at $\sigma = 4$, ResNet50 placed 81.3% of test images above their true severity, DenseNet201 70.9% and the CNN 66.0%, while EfficientNetB0 alone still erred towards lower severity, and then only narrowly, at 38.4% against 34.5%. Apparent robustness of the proliferative DR class under heavy blur reflected the collapse of predictions onto the most severe grade rather than preserved discrimination, indicated by the ResNet50 QWK of 0.0090 at $\sigma = 4$. Grad-CAM was applied to a correct and a misclassified case per architecture across the blur ladder, with attention change quantified as overlap with the unblurred counterpart. That overlap fell as blur increased, and in one case ResNet50 grew more confident while its prediction moved further from the reference grade; with two cases per architecture the evidence is illustrative rather than general. Retinal image quality is therefore an important factor in AI-based DR grading reliability, and image-quality assessment should precede automated prediction, particularly in low-resource screening where acquisition conditions may vary.

I. INTRODUCTION

A. Background

Diabetic retinopathy (DR) is a diabetic-related complication characterized by damage to the retina vasculature that may occur via development of microaneurysms, haemorrhage, exudation, and proliferation of pathological vessels causing permanent loss of sight. Since early-stage disease is often without any symptoms, patients require regular retinal screening in order to get appropriate treatment in time. According to the World Health Organization, the number of people diagnosed with diabetes has increased from 200 million in 1990 to 830 million in 2022, which raises the problem of DR and other complications related to it [2], and, hence, requires continuous NHS diabetic eye screening programs [3].

Fundus photography is the benchmark method for recording the biomarkers that are used to grade DR for severity using manual techniques, which need special skills and are time-consuming. CNNs, on the other hand, can learn automatically the patterns of the retina and classify the severity level based on large datasets of retinas such as EyePACS, APTOS, and Messidor-2 [4, 5]. Transfer learning techniques such as ResNet50, DenseNet, and EfficientNet further improve on this [6, 7].

Practical use relies on the image quality of the input images, since these models depend entirely on the visual patterns that the input images contain. Image acquisition causes the problem through varying illumination, difficulties in acquiring the image, camera problems, and patient factors, with the main one being blurring of the image [8].

Low-cost smartphone fundus cameras extend screening into remote areas where specialist facilities are limited [9, 10], and pairing them with automated grading could reduce screening costs and speed up analysis. Their optics, focusing, lighting, and reliance on manual positioning nonetheless yield lower-quality images than tabletop cameras, with blur the characteristic degradation [9].

Automated grading is therefore developed and validated on clean images yet would be deployed on degraded ones. Architecture comparisons for DR classification exist, and some studies have examined general fundus degradation, but limited

research has systematically quantified the effect of controlled blur levels on five-class DR severity classification while also examining changes in model attention. Rather than restoring degraded images, this research measured robustness as image quality decreased, comparing a baseline CNN with ResNet50, DenseNet201 and EfficientNetB0 under controlled Gaussian blur from $\sigma = 0$ to 4 (0% to 80% blur), with Grad-CAM establishing whether the regions driving predictions remained consistent.

B. Project Aim

The aim of this project was to quantify how controlled Gaussian blur influenced the performance, robustness, and Grad-CAM explanations of deep learning models for five-class diabetic retinopathy severity detection.

C. Objectives

The following objectives were completed:

1. To conduct a critical review of relevant literature in relation to diabetic retinopathy, fundus image analysis, automated classification of DR, degraded image quality, and the robustness of deep learning algorithms.
2. To curate and prepare a standardized fundus image database through the process of preprocessing, balancing, and preparation of experiments.
3. To design and compare four different types of deep learning architectures: baseline CNN and three transfer learning models – ResNet50, DenseNet201, and EfficientNetB0.
4. To train all models using the same experimental protocol on clean fundus images to establish baseline performance.
5. To generate five deterministic Gaussian blur levels and verify the degradation process using image quality measurements, including sharpness analysis.
6. To evaluate model behaviour under increasing blur conditions using classification metrics, including per-class performance, macro F1-score, quadratic weighted kappa (QWK), ROC-AUC, and statistical analysis.
7. To generate Grad-CAM visualisations across different blur levels in order to analyse changes in model attention and interpretability.
8. To critically analyse the results, limitations, and practical implications of deploying AI-based DR grading systems under reduced image quality conditions.

D. Scope

This project focused on five-class diabetic retinopathy severity classification using retinal fundus images. The investigated categories were no DR, mild DR, moderate DR, severe DR, and proliferative DR.

Only one type of image degradation was evaluated: Gaussian blur. Five controlled blur levels were generated to represent progressive image quality reduction, while extreme blur conditions that removed most retinal information were excluded because they were less representative of realistic imaging problems.

The dataset used for this research study was collected from public sources, and there was no requirement for any patient recruitment, clinical trial, or additional ethics approval. As per the limitation in computing capability, only the subset of data was utilized instead of the full

dataset. Hence, the results are a robustness test performed under controlled conditions and not clinical validation.

The project involved an evaluation process rather than development of a new classifier architecture. Techniques like blur-aware learning, image restoration, and domain adaptation were some of the methods that could be used in the future.

E. Research Questions

This research addressed the following questions:

RQ1: What is the impact of gradual increase in Gaussian blur on five-class diabetic retinopathy classification performance using the same held-out fundus images?

RQ2: What diabetic retinopathy classes are most susceptible to blur effects, and are the misclassifications due to under-classification, over-classification, or classification confusion among neighboring levels of disease severity?

RQ3: Does there exist any difference between the baseline CNN, ResNet50, DenseNet201, and EfficientNetB0 models regarding their robustness under increased blur?

RQ4: How does the increase in blur influence the localization and stability of Grad-CAM regions in deep learning-based diabetic retinopathy classification models?

II. RESEARCH AND PLANNING

A. Literature Review

Developing automated systems for screening of DR has recently become a popular research topic, which is motivated by the rising worldwide incidence of diabetes and the difficulties associated with manual retinal image analysis. Computer-aided initial attempts at developing methods of DR screening were based on feature-based image processing, where algorithms were manually constructed by researchers in order to detect certain retinal lesions like microaneurysms, haemorrhages, and exudates. However, such attempts frequently failed since retinal abnormalities can be very different in their morphology, size, position, and intensity. The development of deep learning methods changed the research agenda and allowed building models that could automatically extract visual features from fundus images. Some recent works demonstrated very encouraging results on DR severity classification using CNNs, transfer learning, and explainable artificial intelligence (XAI). However, while performance

improvement using clean datasets has been extensively studied in recent years, there is less work on the influence of image quality degradation on the model's reliability.

1) *Diabetic Retinopathy*: Diabetic retinopathy (DR) is a retinal condition of diabetes that occurs due to the damage in the blood vessels that supply the retina. Exposures to high blood sugar levels may cause damage to the vascular structure, resulting in retinal injury and subsequent vision loss. Diabetic retinopathy is known as one of the main reasons for blindness that could be prevented if detected and treated early enough [5]. This condition is categorized according to the severity of the condition into five levels of disease: no diabetic retinopathy, mild, moderate, severe, and proliferative diabetic retinopathy. Severity is based on the degree of retinal damage, which includes microaneurysm, hemorrhage, exudate,

and proliferation of blood vessels [4]. The severity of the condition is graded using the international clinical diabetic retinopathy severity scale, which gives five grades of diabetic retinopathy from none to proliferative diabetic retinopathy [11].

DR screening in clinical settings is mainly done through retinal fundus photography. The Ophthalmologists observe the retinal pictures taken by a camera and rate their severity levels depending on the pathological symptoms that can be seen in the images. Such screening needs professional expertise and is time-consuming, which makes it hard to conduct on a large scale, especially in areas where there are not enough ophthalmologists [12]. Even though screening has helped in early detection, there are many people with diabetes.

Automated DR grading systems have thus been developed to help doctors through screening process automation and accessibility improvement. Accurate classification into five severity levels has proven difficult, since the distinction between adjacent levels is often quite minor. For instance, differentiating between mild DR and moderate DR requires finding small differences in retinal lesions. Automated systems should consequently learn global patterns of retina and specific disease signs in order to perform correct severity classification.

2) *Fundus Imaging and Quality*: Non-invasive fundus photography captures inner retinal structures, including the optic disc, macula, blood vessels, and the surrounding tissues. In DR detection, these photographs enable the assessment of the progression of the disease using biomarkers like microaneurysm, haemorrhages, hard exudates, cotton wool spots and abnormal vessel growth, which is fundamental for clinical evaluation and deep learning models [12].

Nevertheless, the image quality may differ with imaging technology, environment and the acquisition procedure. Image quality issues often result from blurring due to poor focusing and movement, lack of even illumination, poor contrast and imaging artifacts, which can conceal retinal information and thus complicate the diagnosis process by both clinicians and artificial intelligence models. Liu et al. [8] reported the negative impact of controlled degradations

of the fundus images on CNN-based diagnosis models especially in cases where blurring masked important visual information.

Traditional tabletop fundus cameras generally provide high-resolution retinal images under controlled clinical conditions, but are expensive, require trained operators and are not always practical for remote screening. Smartphone-based and portable cameras offer a lower-cost, more accessible alternative that Iqbal [9] and Chalam et al. [10] identified as having the potential to expand retinal screening through portability and affordability. These devices may, however, produce images with greater variability in quality owing to differences in optical systems, lighting conditions and manual acquisition.

The growing use of portable imaging therefore challenges AI-based DR screening: a model trained only on high-quality clinical images may not maintain that performance on lower-quality images captured in real-world environments. Evaluating how image quality degradation affects deep learning-based DR grading is necessary for understanding the practical reliability of automated screening.

3) *AI-based Diabetic Retinopathy Severity Grading*: Automated DR grading has moved from handcrafted-feature pipelines towards deep learning. Earlier computer-aided diagnosis systems detected abnormalities such as microaneurysms, haemorrhages and exudates, but depended on manually selected features and so generalised poorly across datasets and imaging conditions. Convolutional neural networks (CNNs) learn hierarchical features directly from retinal images, suiting fundus analysis because abnormalities appear as spatial patterns, textures, shapes and colours.

Transfer learning is standard because medical datasets are small by natural-image standards, so ImageNet-pretrained models reuse general visual features. Reported clean-image performances are inconsistent: Qureshi et al. [4] scored 98% in five-level classification on EyePACS while the use of an active deep learning model needed a complicated training process, showing that there is a compromise between accuracy and practical implementation. Vaish et al. [6] used improved ResNet50 on EyePACS and APTOS and attained 82.37% and 87.35% respectively, and the application of ResNet50 in the area still makes sense [13]. Mehboob et al. [5] achieved 83.78% from CNN ensembles on EyePACS but the scores were influenced by imbalanced data and different classes of severity. EfficientNet family considers accuracy versus computational efficiency: Matthew et al. [14] recorded 91.85% from EfficientNetB0 in a mobile version, and Moodu et al. [7] applied EfficientNet for retinal images.

Severity level classification into five classes is more difficult compared to detection, which involves just classifying disease attributes as present or absent. It becomes necessary to differentiate adjacent severity levels, where for instance both moderate and mild DR can exhibit similar visual changes. Issues such as class imbalance and similarity between adjacent classes prevail.

Public datasets serve the purpose of enabling reproducibility and comparisons between studies: EyePACS offers the large set, whereas APTOS and Messidor-2 offer distinct imaging features; these have been widely employed in preprocessing, transfer learning and model optimization research [15]. Comparing directly is still hard due to different preprocessing techniques, dataset splits, evaluation metrics and class balancing methods.

Explanation is needed since deep learning models are generally viewed as black-box systems, and users want assurance that prediction is based on relevant retinal regions and not unrelated features of the image. Muntaha et al. [16] managed to achieve 95.25% accuracy while utilizing Grad-CAM to visualize regions responsible for making the predictions.

Still, most of this research has been conducted on high-quality fundus images, and improvement has been obtained via architectural tweaks, preprocessing and training techniques. Little attention has been paid to whether such models will function in cases of degraded images, and this might be important in actual deployment on portable fundus cameras.

4) *Existing Deep Learning Models for Blur-Robust Image Classification*: A potential weakness of any classification system is its ability to work properly under degraded inputs, because an image received by the model during operation might have different quality than the images used for training, since

motion, defocusing, illumination changes and sensor limits would reduce its quality and influence the prediction results. That does not apply only to retinal images, because convolutional neural network benchmarking on images corrupted by various factors has revealed that accuracy decreases along with quality of the input [17], and blur and noise are worse degrading factors compared to compression and contrast changes [18, 19].

There are very few works studying blur resistance for grading of diabetic retinopathy severity, since most of them aim at achieving higher accuracy when working with fundus images of high quality rather than studying blur degradation of diagnostic ability of CNNs. Liu et al. [8] is the nearest experiment in terms of analysis of degradations of fundus image, including blur, lighting interference and retinal artefacts in their effect on CNN-based diagnoses using DenseNet and ResNet architectures.

That precedent evaluated existing models under controlled degradation rather than restoring low quality images, and the same idea holds true here – modeling the behavior of models under degradation gives an understanding about the robustness of the classification method.

Robustness has been observed that the robustness of models has been investigated in fields like self-driving cars, surveillance and object detection, and models that have been trained using mostly clear images often perform poorly in case of degraded images. However, augmentation techniques and other such methods are not always applicable in the case of medical imaging as

the abnormalities present in medical images are usually small and get washed out during the process of degradation.

Blur is a more difficult challenge as there are features that are at different spatial resolutions – some larger ones like blood vessels will still be clearly visible while smaller ones like microaneurysms are hard to distinguish under degradation.

Previous research did not look into the relationship between blur magnitude, performance of DR grading and model interpretability, as they either considered image classification without any blur or looked into degradation without five-class DR classification and Explainable AI; no prior work has been done to compare different architectures under the same level of blur.

The current project looked at the robustness of models instead of image de-blurring, and the aim was to understand how a baseline CNN and three different transfer learning models performed with increasing Gaussian blur, allowing for comparison of different models.

5) *Research Gap:* Deep learning models have demonstrated encouraging results in automated diabetic retinopathy severity grading, where the use of CNN, transfer learning, and ensemble approaches has helped improve classification of fundus images based on their severity levels [4, 5]. However, existing studies are predominantly focused on increasing the accuracy of deep learning algorithms using high-quality datasets.

Fundus photography on portable devices and smartphones increases DR testing opportunities but can create image blurring, uneven lighting and contrast problems [9, 10]. Despite the proven influence of image quality degradation on CNN-based disease diagnosis [8], there is little research dedicated to investigating the effect of controlled blurring on severity level

classification in five classes by different types of deep learning algorithms.

Additionally, robustness studies often neglect the combination of the evaluation of performance with an analysis of explainability. A reduction in classification accuracy does not necessarily indicate that deep learning models cannot recognize retinal features due to blurring and need to allocate attention to other elements instead [16].

The problem identified through this literature review, and to which this project contributes a solution, is the impact of increasing blurriness in images on the task of five-class DR grading severity through deep learning algorithms, along with accompanying Grad-CAM analyses. The research question that this project explores is how Gaussian blur affects the performance, robustness and visualization of CNN-based DR grading severity.

Three concepts from the literature review were incorporated into the study design: the need for robust feature learning due to subtle differences in DR severity, image quality is an important factor to consider when assessing the performance of any AI for clinical screening, and combining qualitative and explainable AI provides more understanding of the model behavior.

B. Methodology

The methodology was aimed at studying the effect of degradation of fundus images on automated classification of the severity of diabetic retinopathy (DR). Rather than proposing a new architecture for classification, the current study concentrated on assessing the robustness of existing models in the setting of controlled image degradation. In total, there were six stages of the experiment: dataset selection, dataset preparation, model training, blur introduction, performance assessment, and explainable AI analysis. To begin with, the unified dataset of DR fundus images, combining multiple public datasets, was selected and prepared. Then, the prepared dataset was split into training, validation, and test sets. Four classification models were trained on the images without any degradations and were evaluated on the five blur levels introduced to the images in a controlled manner. Training and testing of models was conducted in TensorFlow/Keras framework, while evaluation was performed by means of Python scientific computing tools, such as scikit-learn [20, 21]. Classification performance and visual attention changes were assessed based on quantitative evaluation metrics and Grad-CAM explanations. The experimental pipeline is presented in Figure 1.

1) *Dataset Description:* Unified diabetic retinopathy dataset publicly available on Kaggle was chosen by combining the images from popular retinal datasets such as EyePACS, APTOS and Messidor. The dataset consisted of 92,501 anonymised fundus images that were annotated based on the five clinical categories of DR severity, including no DR, mild DR, moderate DR, severe DR and proliferative DR. EyePACS was created as a flexible telemedicine screening test [22], whereas the Messidor dataset was made publicly available as a reference database for retinal images analysis [23].

This dataset was chosen due to the fact that publicly available benchmark datasets are crucial for the reproducibility of

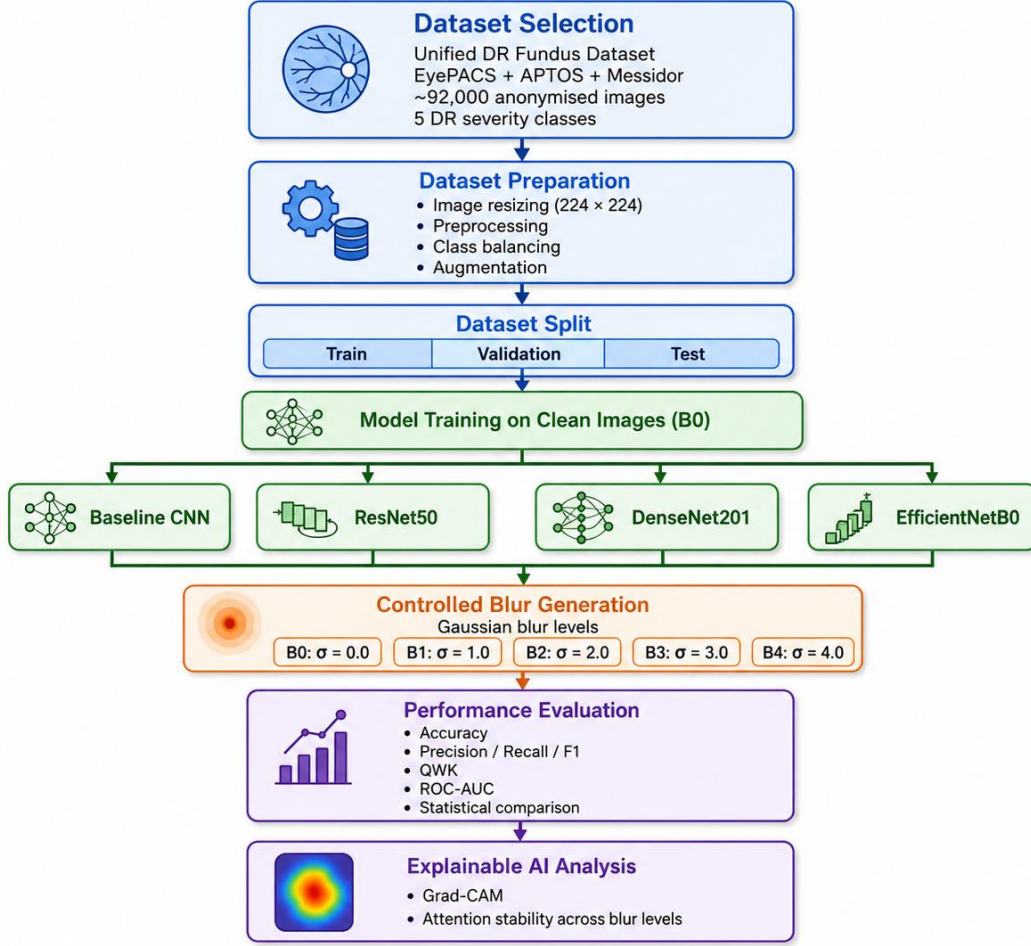


Fig. 1. Overview of the experimental methodology pipeline used to quantify the influence of fundus image quality degradation on automated diabetic retinopathy severity detection.

research and a proper comparison of it with previous studies in this area. EyePACS, APTOS and Messidor datasets have been widely used in literature in the evaluation of automated DR grading algorithms [4, 5, 15]. Additionally, the dataset from multiple sources gives more diversity in image characteristics than the single one.

However, the dataset had a large class imbalance issue, where normal and mild cases were more in numbers than the severe cases. This is a common problem in DR classification as there will always be a smaller number of samples at the stage of severe cases. So, the techniques for dataset creation were used to make a more balanced training environment.

The dataset used for this project was from the secondary sources which have been already publicly available and anonymous. No sample recruitment or any kind of intervention has been done. The project followed the university research ethics approval process for the use of secondary healthcare datasets.

2) *Choosing the Models*: The chosen deep learning models were chosen because of their proven performance in image classification problems, especially in medical image classification and retinal image classification. It is important to note that the aim of choosing the models is not only to choose

the highest performing architectures but to compare different architectures in terms of robustness against blur according to feature extraction differences. The chosen models consist of one baseline CNN architecture trained from scratch and three transfer learning models: ResNet50, DenseNet201, and EfficientNetB0.

a) *Baseline CNN*: A simple CNN architecture was used as the baseline reference, since there needed to be a comparison model for measuring the added value of the deeper and pretrained

architectures. The baseline CNN architecture was made up of four convolutional layers that were meant to learn hierarchical visual features from the fundus images.

This model was not meant to deliver the best classification performance. Its goal was to act as an easy to interpret reference point that would show the difference in feature learning and robustness brought about by more complex architectures. Comparing pretrained models against a basic CNN allowed the effect of model complexity to be evaluated.

b) *Transfer Learning Models*: Three different classification models that are pre-trained were chosen since previous studies proved their efficiency for medical images and DR classification tasks. ResNet50 adds residual connections enabling

training of deeper networks without the degradation problem that occurs during training of equivalently deep but not residual models [24]; DenseNet201 uses dense connectivity in which each layer of the block is directly connected to all following layers, and therefore provides more recent feature maps [25]; and finally, EfficientNetB0 scales the network in width, depth and resolution taking into account number of parameters [26]. They were all pre-trained on ImageNet dataset [27].

ResNet50 was chosen due to the fact that residual learning became widely used for DR classification tasks and fundus image analysis. The idea of residual connections enables deeper networks to learn complex representations and overcome optimization problems. Previous works prove the ability of the architectures based on ResNet to perform well on EyePACS and other retinal datasets [4, 6].

The model chosen was EfficientNetB0 since it offers a balance between efficiency and classification performance. Such a trait is vital for medical applications in which a machine learning model might have to be run on constrained devices. It has been used before to classify DR and mobile-based medical AI systems in earlier studies [14, 7].

DenseNet201 was chosen based on the earlier research on degraded images, indicating that densely connected networks are better in preserving features under difficult conditions. In contrast to ResNet, DenseNet promotes feature reuse in that every layer connects to other layers. Therefore, DenseNet201 was chosen especially for the purpose of investigating the impact of its feature propagation behavior in blurred fundus images [8].

3) *Evaluation Metrics and Statistical Comparison:* The value of accuracy alone can be misleading in assessing the performance of DR classification models, as the retinal datasets have unbalanced class distributions, with a

relatively higher amount of normal and early-stage images than the images corresponding to more severe forms of DR. A classifier could obtain very high values of overall accuracy and low values for particular minority severity classes. Hence, several performance metrics have been used to assess the model performance, including precision, recall, F1-score, quadratic weighted kappa (QWK), and ROC-AUC.

Per-class precision, recall, and F1-score were employed to investigate which particular DR severity levels are impacted by the blur effect. Macro F1-score was chosen since it treats all classes equally irrespective of their frequency. QWK was considered due to the fact that DR grading is a classification problem, in which predicting an adjacent severity level is not the same as predicting a wrong grade. ROC-AUC was selected to measure the ability of classifiers to separate classes from each other.

A statistical comparison has been done to investigate whether the obtained performance differences are statistically significant. The complete metric definitions and statistical testing procedure are described in the Practical Work section.

4) *Explainable AI with Grad-CAM:* Deep learning algorithms tend to be regarded as black-box algorithms because it is hard for humans to understand the reasoning behind the prediction. Understanding how the algorithm makes its predictions is important in medicine because clinicians need

assurance that the algorithm is paying attention to medically relevant anatomical features instead of other image features.

Gradient-Weighted Class Activation Mapping (Grad-CAM) was used in the experiment. Grad-CAM is a technique which marks regions of an image that contribute the most to the CNN prediction [28]. The use of Grad-CAM for fundus image analysis is justified because it can help visually check whether models pay attention to clinically significant areas of the retina.

Previous studies on DR have used Grad-CAM in order to increase the transparency of automated retinal grading systems. Muntaha et al. [16] have shown the usefulness of Grad-CAM for highlighting clinically significant regions of the retina involved in the decision-making process during DR classification. In this project, Grad-CAM has been extended to the analysis of progressive blurring of the fundus images.

Therefore, Grad-CAM provided an additional perspective alongside numerical evaluation metrics by showing how image degradation influenced not only classification performance but also model visual reasoning.

III. PRACTICAL WORK

A. Exploratory Data Analysis

The five levels of diabetic retinopathy (DR) severities (No DR, Mild, Moderate, Severe and Proliferative) are very much imbalanced prior to the sampling process (Figure 2): No DR comprises about three-quarters of the training dataset, with the validation and test datasets exhibiting the same imbalance. However, the data imbalance can create a bias towards the more dominant classes in the case of classifiers; hence, class distribution is one of the controlled factors.

Due to resource constraints, where there are five levels of blurring and four architectures, a balanced sample size will be used for modeling (Figure 6), as outlined in Section 3.2.1. The number of samples per class is restricted, where there are less number of samples of severe DR due to scarcity; validation and test datasets are kept separate such that architectures can be compared without the influence of class distribution.

Since the experiment involved manipulating blur, the base images needed to be proven as containing visually perceptible information through two criteria. Variance of the Laplacian is a non-reference criterion of the edge sharpness of an image, where a high value would mean a sharper image [29]. This proves that the original images were sharp prior to manipulation. Peak signal to noise ratio (PSNR) is a full-reference criterion comparing a manipulated image to its own original [30], establishing that the manipulation was ordered and controlled.

The median for unmodified images with $\sigma = 0$ was 342.12. Both statistics were calculated on the entire padded image. No mask was used, meaning that the circular retina and the black border around it were both included in all calculations, each of which comes with its own distinct implication.

With respect to the variance of the Laplacian statistic, the black border is a large area of near-zero variance, meaning that its inclusion tends to deflate the reported statistic. On the other hand, the difference between the black border and the illuminated retina is an edge of high variance, meaning that the Laplacian response of the edge is large; the fact that this edge is

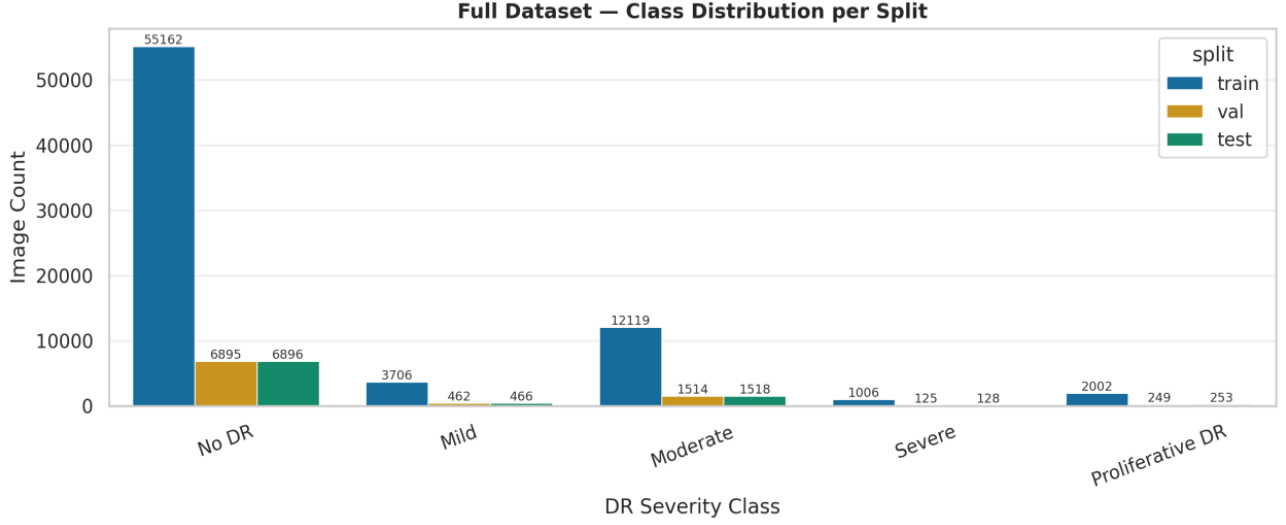


Fig. 2. Full dataset class distribution across training, validation, and testing splits.

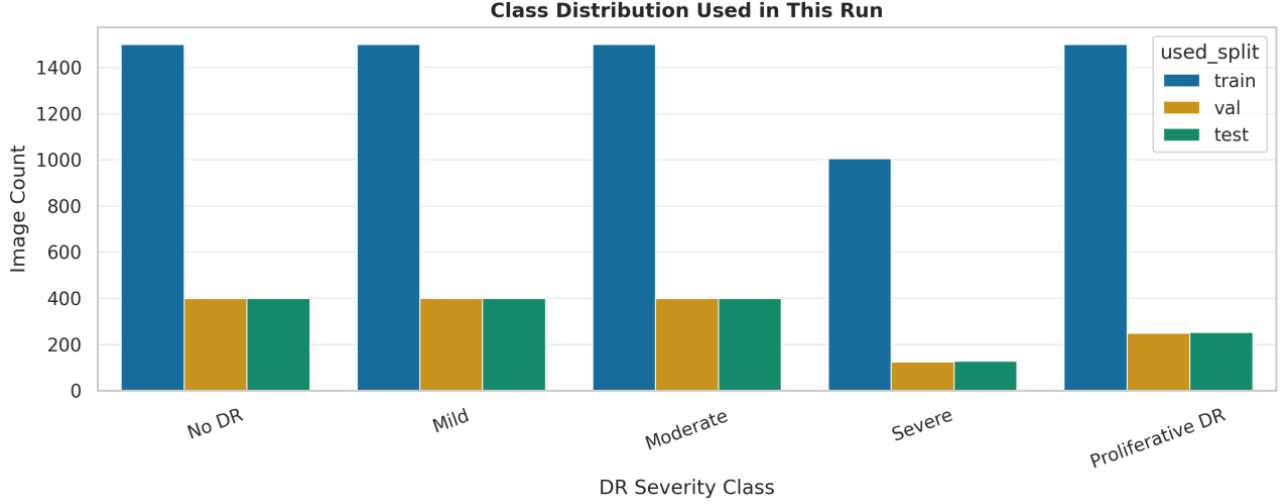


Fig. 3. Class distribution of the dataset subset used for model training and evaluation.

always the same means that it tends to inflate the variance of the Laplacian statistic by some unknown but constant amount. There is no ratio at which these two opposing biases will cancel out, which means that 342.12 is not the sharpness of the retinal tissue.

For PSNR, the bias is clear-cut: blurring of a uniformly black area will leave it uniform, and the contribution to the mean from the background is thus negligible while still contributing to the count. This means that the background systematically inflates the measured quantity, and all values reported here should be considered as upper bounds for the retina-only values.

The bias does not influence the level comparison, however: The frame structure is the same for each σ and image, meaning the same bias applies to each measurement. The order and separation of the levels remains intact; the claim that can be justified is only that the five levels are ordered and well-separated; the claim that the retina had an absolute sharpness of 342.12 cannot be justified, and re-calculations with masking

will be required before reporting any such absolute number.

In Fig. 3, sharpness values are reported for the five levels, with the range of minima and maxima found in A.4; the median decreases monotonically with increasing σ , and ranges for adjacent levels above $\sigma = 1$ are small.

PSNR was determined for every degraded image relative to the undistorted version of itself; the undistorted point is a stand-in, not an actual measure, because there is no difference between the two images being compared. Decibel-based, PSNR increases with similarity to the original and thus the opposite direction of Laplacian variance. Figure 4 depicts the monotonic decrease in PSNR with increasing σ , proving actual degradation instead of random corruption.

The 30 dB mark in Figure 4 is a convention, not a measured characteristic: conventionally, for 8-bit images, compression and decompression has reported 30 to 50 dB [30]. It is an engineering heuristic, not a scientifically validated perception threshold, which does not have any clinical validation in retinal

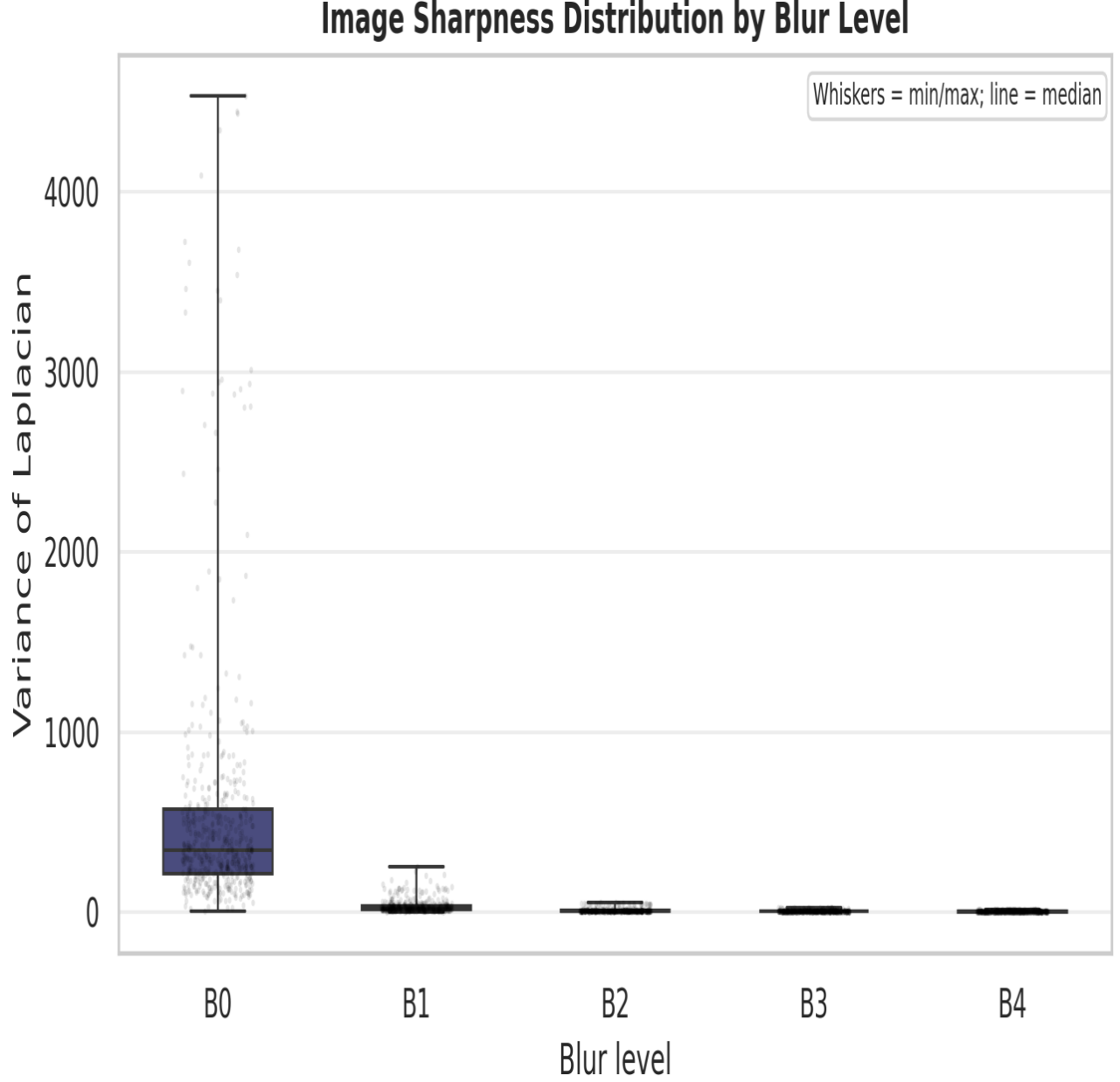


Fig. 4. Median image sharpness measured using variance of Laplacian across different blur levels.

photography, nor any diagnostic relevance to images above or below that mark.

In Figure 5, it is increasingly difficult to discern blood vessels, the margins of optic discs, and lesion-like areas as the σ goes from 0 to 4, mimicking substandard retinal imaging like poor smartphone cameras or improper image acquisition.

Before training, all images were confirmed to contain three colour channels, cropped to remove black borders while retaining the circular retinal region, and resized to 224 x 224 pixels.

B. Dataset Preparation and Experimental Setup

After the exploratory analysis had been completed, the dataset and experimental pipeline were prepared so that each model could be evaluated under the same conditions. This was important because the purpose of the project was to compare how different architectures responded to image degradation rather than to maximise the performance of any individual model. The training, validation, preprocessing and blur-evaluation procedures were therefore kept consistent across the four models wherever their architecture allowed it. The dataset used in this project was obtained from the publicly available EyePACS-Aptos-Messidor diabetic retinopathy unified dataset on Kaggle [1].

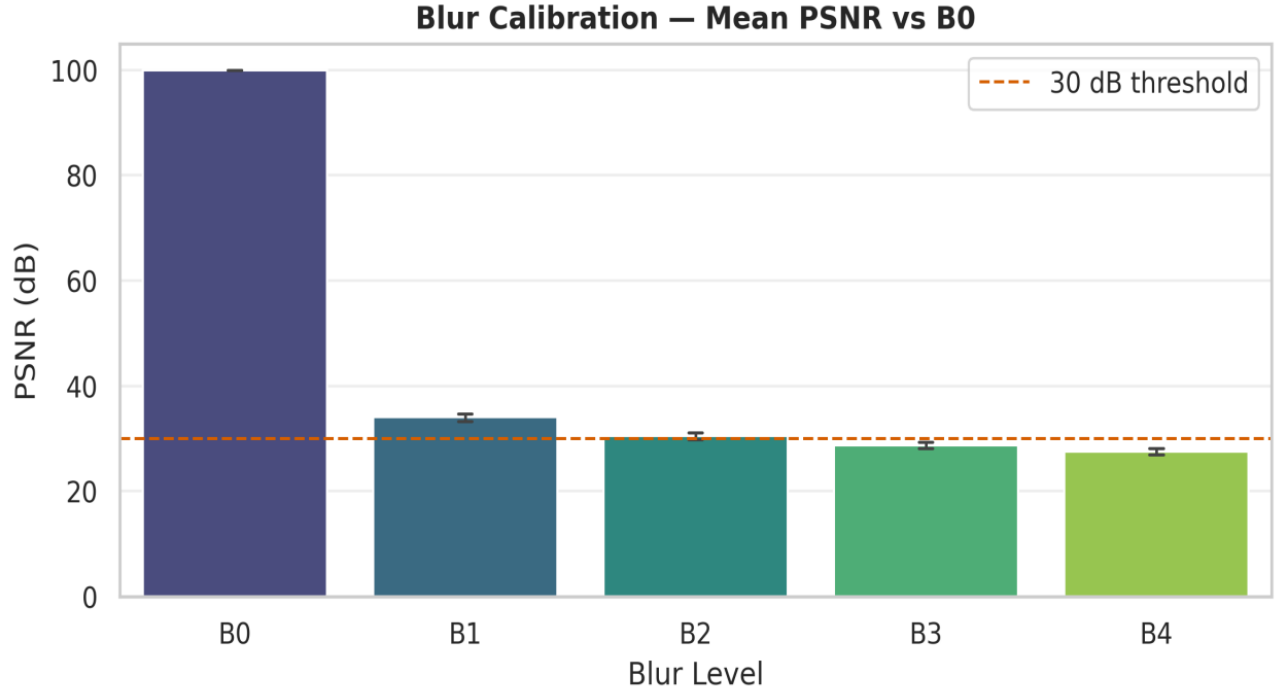


Fig. 5. Blur calibration using mean PSNR across five blur levels.

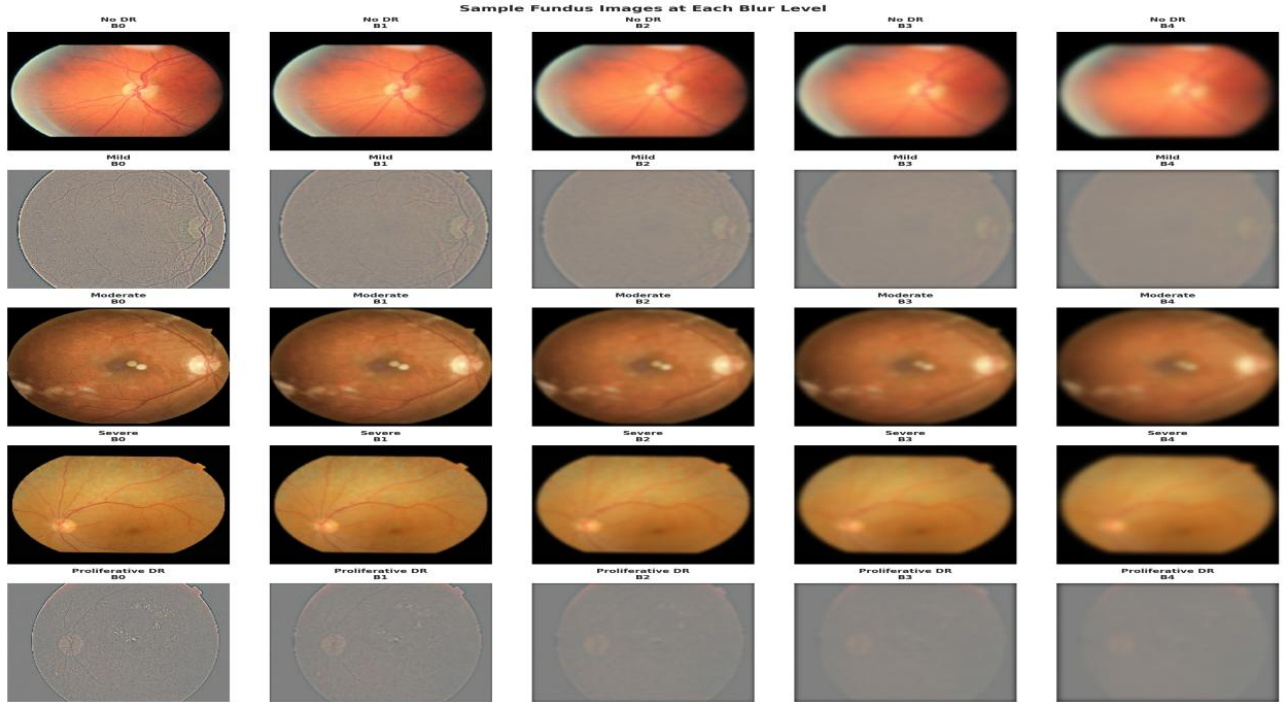


Fig. 6. Example fundus images showing progressive degradation from clean images ($\sigma = 0$) to heavily blurred images ($\sigma = 4$).

1) *Dataset Sampling and Experimental Subset*: The unified dataset contained 92,501 retinal fundus images across the training, validation and test splits. Training every architecture repeatedly on the full dataset was not practical within

the free Kaggle GPU allowance, particularly because each trained model also had to be evaluated across five blur levels.

FAST_MODE was therefore enabled and a fixed per-class sampling strategy was used.

The training subset was limited to a maximum of 1,500 images per class, and the validation and test subsets to a maximum of 400 images per class. Where fewer images existed, all available images were retained. This produced 7,006 training

images, 1,574 validation images and 1,581 test images. The final training support was 1,500 No DR, 1,500 Mild, 1,500 Moderate, 1,006 Severe and 1,500 Proliferative DR images. The Severe class remained smaller because only 1,006 images were available in the original training split.

For Python, NumPy, TensorFlow, and data set sampling, a fixed random seed of 42 was used. This was to ensure that the same subset and shuffling was achieved in case the code was run again. Since only a sample is being taken, the resulting absolute values of the performance will be indicative and not population values. Every network has been evaluated on the same sample, making relative comparisons valid for robustness analysis.

2) *Best-Effort Patient Leakage Check*: A potential issue with retinal datasets is the fact that the same patients' images may appear in different splits. In case this issue occurs, the results of the study may be overestimated because of the ability of the model to see similar retinal characteristics both in the training process and the testing process.

The unified dataset did not offer an additional verified table of patients' identities. Hence, a leakage check was done based on patient ID generation from each image file name. As a result of this check, there were found 340 overlapping IDs in between training and validation, 319 between training and testing, and 75 between validation and testing.

These values were not used as leakage information because the ID was based on the image file name and not on the clinical patient ID. Instead, it was considered a limitation and proved that it is impossible to guarantee the full independence of patients and their images.

3) *Image Preprocessing*: Input images were preprocessed due to variability in terms of retinal framing and dimensions of images in the source dataset. First, black borders outside the retinal image were cropped based on intensity. To preserve useful information from the peripheral regions, a margin was kept around the detected region of retina.

In preprocessing, padded resize technique was used instead of stretch. Each retinal image after cropping was resized with padding to 224×224 pixels. Size 224×224 pixels was selected as it was compatible with all the four models and minimized computational expense.

After resizing, the preprocessing expected by each architecture was applied. The baseline CNN used pixel scaling to the range required by the custom network, while ResNet50, DenseNet201 and EfficientNetB0 used their corresponding Keras application preprocessing functions. These architecture-specific input transformations were retained because pretrained ImageNet models expect the same preprocessing used during their original training.

The experimental pipeline was implemented in Python using TensorFlow/Keras for model construction and training and scikit-learn for class weighting and evaluation metrics. The software environment and exact package versions were recorded by the executable notebook so that the experiment could be reproduced from the submitted project artefacts.

4) *Training Augmentation and Class Imbalance*: Augmentation was applied only to the training data because validation and test images needed to remain unchanged for fair measurement.

TABLE I
DEFINITION OF THE FIVE BLUR CONDITIONS, SHOWING GAUSSIAN σ AND THE CORRESPONDING DISPLAY PERCENTAGE.

Blur level	Display level	Gaussian σ	Meaning
$\sigma = 0$	0%	0.0	Clean baseline
$\sigma = 1$	20%	1.0	Weak blur
$\sigma = 2$	40%	2.0	Mild blur
$\sigma = 3$	60%	3.0	Moderate blur
$\sigma = 4$	80%	4.0	Strong blur

Among flip operations, horizontal flipping alone was retained because left and right retinal images can reasonably be mirrored. Vertical flipping was rejected because an upside-down retinal orientation is not anatomically realistic and could teach the model patterns that would not normally occur during fundus acquisition.

The final data augmentation process also consisted of small random rotations, zooming, and translations. They were limited to be mild enough to provide natural visual variability, but not strong enough to create a purposeful blur, which is our target condition.

The issue of class imbalance persisted even after per-class sampling due to the relatively small number of samples in the Severe class. To account for this, the balanced class weights were computed based on the labels of the training set and incorporated into the model training process. The training loss function was modified to use weighted categorical cross-entropy as opposed to a more complex focal-loss-based stack, which was previously taken into consideration. It was necessary to limit the number of experimental variables, since the question was focused only on image blur and architectural stability.

5) *Controlled Blur Generation*: Deterministic Gaussian blur was used as a controlled degradation variable that could be increased systematically while keeping the underlying retinal image unchanged. The five conditions are defined in Table 1.

comparison. After supervision, 80% blur became the practical limit, where the lesions and anatomical structures become heavily blurred but the image remains recognizable as a fundus image.

Gaussian sigmas had initially been calibrated to images that contained roughly 300 pixels but were scaled down to 224×224 pixels. This approach was used throughout the experiments due to consistency requirements, although the inconsistency is clearly stated as one of the limitations.

The experiment employed the clean training / test degradation paradigm, where the models were trained and validated solely on images of $\sigma = 0$, while blurring is applied to test copies of images. Thus, this experiment evaluated the response of the model trained under normal conditions to the degradation of an image at the test time.

Another design approach that included blur as part of the training process was evaluated. This might have contributed to better robustness, but this design would be answering a different question, which is how to reduce blur instead of the effect of blur on the existing classifier. Clean training was thus chosen, while blur training is left for future research.

Degradation was confirmed by visualization by means of creating a montage ranging from $\sigma = 0$ to 4 (0% to 80% blur) and also by calculating PSNR and variance of the Laplacian.

Median Laplacian variance fell from 342.12 at $\sigma = 0$ down to 21.48, 5.32, 3.21 and 2.56 at $\sigma = 1$ to 4, confirming a strong and ordered loss of high-frequency detail.

6) *Model Training Protocol*: A single configuration was applied to all four architectures, a design decision rather than an oversight. Input size was 224 x 224, batch size 32, dense-head dropout 0.35, and optimisation used Adam with clipnorm 1.0. The simple CNN was trained from scratch as a four-block reference.

Selection was informed by a limited pilot comparison rather than an exhaustive grid or random search: candidate settings were trialled for validation behaviour under the Kaggle session limits and the best-behaving configuration retained.

The independent variables were architecture and blur severity; tuning each network separately would have introduced optimisation effort as a third, uncontrolled variable, leaving differences between architectures, and between levels of σ , no longer attributable to architecture or degradation alone. Holding the configuration constant preserved that attribution, at the cost of per-model optimality.

224 x 224 is the resolution at which the ImageNet weights for ResNet50, DenseNet201 and EfficientNetB0 were trained [27]; preserving it avoided resampling those filters away from their learned scale, and gave all four models identical pixel data. The batch size of 32 was the largest fitting the free Kaggle GPU allowance for the largest network. Adam was used with its published default moment decay rates [31]. The two-phase schedule follows the established result that lower convolutional layers encode transferable general features while upper layers become task-specific [32]. Batch-normalisation layers were held frozen during fine-tuning because recomputing batch statistics on a small resampled training set would shift the normalisation against which the pretrained weights were fitted [33].

For the three pretrained models, Phase 1 froze the backbone and trained the classification head for up to 10 epochs at 1 x 10⁻³. Phase 2 unfroze the upper 50% of each backbone and fine-tuned for up to 20 further epochs at 2 x 10⁻⁵, the lower 50% remaining frozen to retain general pretrained features. The baseline CNN received head training only.

Early stopping on validation loss restored the best weights, and ReduceLROnPlateau cut the rate by a factor of 0.3 after two epochs without validation-loss improvement, with a floor of 1 x 10⁻⁷. Because the search was a pilot, the absolute performance reported in Section 3.3 should be read as a lower bound for every architecture, and the relative ranking of the four models could change under per-model tuning. A systematic search over learning rate, batch size and unfreezing depth on the validation split at $\sigma = 0$, re-evaluated across the full blur ladder, is the first item for further work.

7) *Evaluation and Statistical Procedure*: Accuracy alone was insufficient for an imbalanced, ordinal five-class problem. Each model and blur level was evaluated on accuracy, balanced accuracy, macro precision, recall and F1-

score, quadratic weighted kappa (QWK), mean absolute grade error, off-by-one accuracy, under-grading rate and over-grading rate, with one-vs-rest macro and micro ROC-AUC where class probabilities were available.

Macro F1 weighted each severity class equally; QWK reflected the ordered structure of DR grades. Per-class sensitivity, specificity, precision and F1-score were recorded at every blur level, tracking class-level deterioration across the ladder. Confusion matrices were rendered at $\sigma = 0, 2$ and 4.

Two exploratory statistical comparisons were performed at alpha = 0.05. One-way ANOVA compared class-level metrics across the four models at $\sigma = 0$, and across $\sigma = 0$ to 4 (0% to 80% blur) within each model. Following supervision the analysis was expanded from sensitivity alone to sensitivity, specificity, precision and F1-score, with a Tukey post hoc on significant omnibus tests.

The tests were conducted carefully, as each group contained only five observations at the class level, and multiple comparison adjustment was made for only the Tukey pairwise set, not the four dependent variables or per model tests. These limitations are stated explicitly rather than reading non-significant p-values as evidence of no blur effect.

Grad-CAM visualisations covered two cases per architecture, one correct and one misclassified on clean input, across $\sigma = 0$ to 4. Holding each case constant allowed attention to be compared as only sharpness changed, the shift quantified by overlap with each map's unblurred counterpart. All outputs are presented in Section 3.3.

C. Experimental Results

This section reports the experimental outputs obtained from the four models under clean and progressively blurred test conditions. Results are presented factually using classification metrics, degradation curves, per-class performance, ordinal error analysis, statistical tests, Grad-CAM visualisations, and computational cost. Interpretation of these results is provided separately in Chapter 4.

1) *Training Behaviour*: The training and validation curves were recorded for the baseline CNN, ResNet50, DenseNet201, and EfficientNetB0. For the pretrained models, the curves reflected the two-stage training procedure, consisting of head training followed by fine-tuning of the upper part of the pretrained backbone. The curves are shown in Figure 7.

The baseline CNN contained 423,877 parameters and required approximately 22.42 minutes to train. ResNet50 contained approximately 24.1 million parameters and required 44.99 minutes.

DenseNet201 contained approximately 18.8 million parameters and required 58.84 minutes, while EfficientNetB0 contained approximately 4.38 million parameters and required 50.36 minutes.

The curves also show the epoch at which the best validation weights were retained through early stopping.

2) *Clean-Image Performance at $\sigma = 0$* : All models were evaluated on the unmodified test set of 1,581 images (Table 2). The three pretrained models are separated by less than 0.04 in macro F1 and less than 0.01 in QWK, margins narrower than the within-model spread between severity classes, so architecture choice was not the dominant factor at $\sigma = 0$, and no single architecture led on every metric. The base CNN ranks below all three in each category, having 0.29 accuracy and 0.22 macro

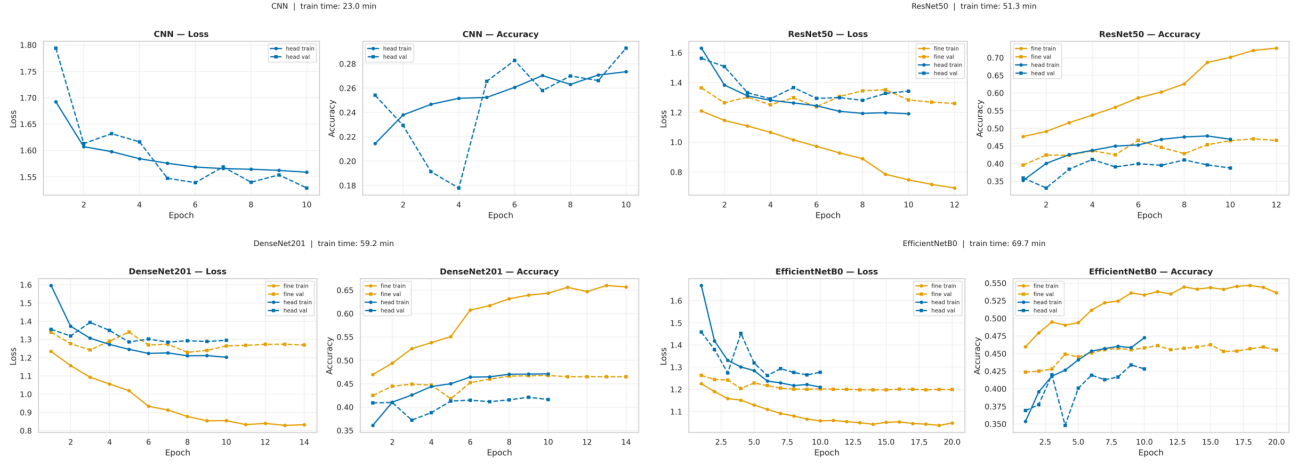


Fig. 7. Training and validation loss curves for CNN, ResNet50, DenseNet201, and EfficientNetB0.

TABLE II
CLEAN-IMAGE PERFORMANCE OF THE FOUR MODELS AT $\sigma = 0$.

Model	Macro F1	QWK	Macro ROC-AUC
CNN	0.2211	0.2076	0.6022
ResNet50	0.4951	0.6804	0.8109
DenseNet201	0.4564	0.6861	0.8093
EfficientNetB0	0.4716	0.6881	0.8046

TABLE III
MACRO- AND MICRO-AVERAGED ONE-VS-REST ROC-AUC BY MODEL AT $\sigma = 0$.

Model	Macro ROC-AUC	Micro ROC-AUC
CNN	0.6022	0.6441
ResNet50	0.8109	0.8201
DenseNet201	0.8093	0.8154
EfficientNetB0	0.8046	0.8154

F1, consistent with all predictions converging on the majority class: Recall 0.71 for No DR vs 0.11 for Mild and 0.02 for Moderate DR.

All three pre-trained networks exhibit the same class-wise trend (see Figure 8): Moderate DR is weakest, having F1 0.22 – 0.35, while Proliferative DR is strongest, with F1 0.63 – 0.71. Such trend is consistent with Moderate DR being identified through the presence of small lesions that take up little space at the image resolution used, while Proliferative DR features more extensive changes in the vasculature.

Liu et al. [8] report QWK from 0.6977 for AlexNet to 0.8056 for ResNet-50 on EyePACS, the source of the images used here. The QWK here, 0.6804 to 0.6881 across the three pretrained models, falls just below that range, the expected outcome for a 7,006-image training subset of the same pool, not the full set. The comparison is confined to QWK: the F1 of Liu et al. [8] is weighted rather than macro, dominated by the largest classes and not comparable with the macro F1 in Table 2; and they did not evaluate EfficientNet, so the EfficientNetB0 results here extend their comparison rather than reproduce it.

3) *ROC-AUC on Clean Images*: One-vs-rest ROC-AUC was computed at $\sigma = 0$. The macro- and micro-averaged values are given in Table 3; per-class ROC-AUC was not among the metrics retained, although per-class sensitivity, specificity, precision and F1-score were recorded at every blur level. The curves for the pretrained models are shown in Figure 9.

The three pretrained models are separated by less than 0.02 in macro ROC-AUC, from 0.8046 for EfficientNetB0 to 0.8109 for ResNet50, so this threshold-free measure reproduces the narrow architectural spread reported in Section 3.3.2. The CNN’s macro value of 0.6022 and micro value of 0.6441 sit close to the 0.5 expected of random ranking, so its ordering of

cases by severity is near-chance rather than weak but genuine discrimination. Because the per-class

values were not retained, class-level separability is described from the per-class F1 values in Section 3.3.2 rather than from per-class ROC-AUC.

4) *Performance Across Blur Levels*: Degradation from $\sigma = 0$ to 4 was measured on the same held-out test images at every blur level. Macro F1 is given in Table 4, quadratic weighted kappa in Table 5, and the degradation curves in Figure 10.

Averaged across the three pretrained models, macro F1 fell by 67.5% and QWK by 88.1% between $\sigma = 0$ and 4; with the baseline CNN included, the corresponding averages are 61.8% and 83.9%. For the same three models, the mean step-wise change in macro F1 was -7.6% for $\sigma = 0$ to 1, -35.8% for 1 to 2, -37.2% for 2 to 3 and -12.9% for 3 to 4. Degradation was therefore not linear in σ : the two largest single-step losses both fall between $\sigma = 1$ and 3, that is between 20% and 60% blur, while the first step is comparatively cheap and the final step adds little because performance is already close to the floor.

The baseline CNN is excluded from these averages for two reasons: it improves from $\sigma = 0$ to 1 on macro F1 and QWK while accuracy and both ROC-AUC values fall, and its macro F1 at $\sigma = 4$, 0.1656, lies below the approximate chance value of 0.193 for this test support, leaving its step-wise changes without a usable interpretation.

The position of this knee converts the image-quality gate recommended in Chapter 5 from a general statement into a threshold derived from the measurements reported here: a gate set at or before $\sigma = 1$ retains most of the clean-image performance, whereas one set at $\sigma = 3$ does not.

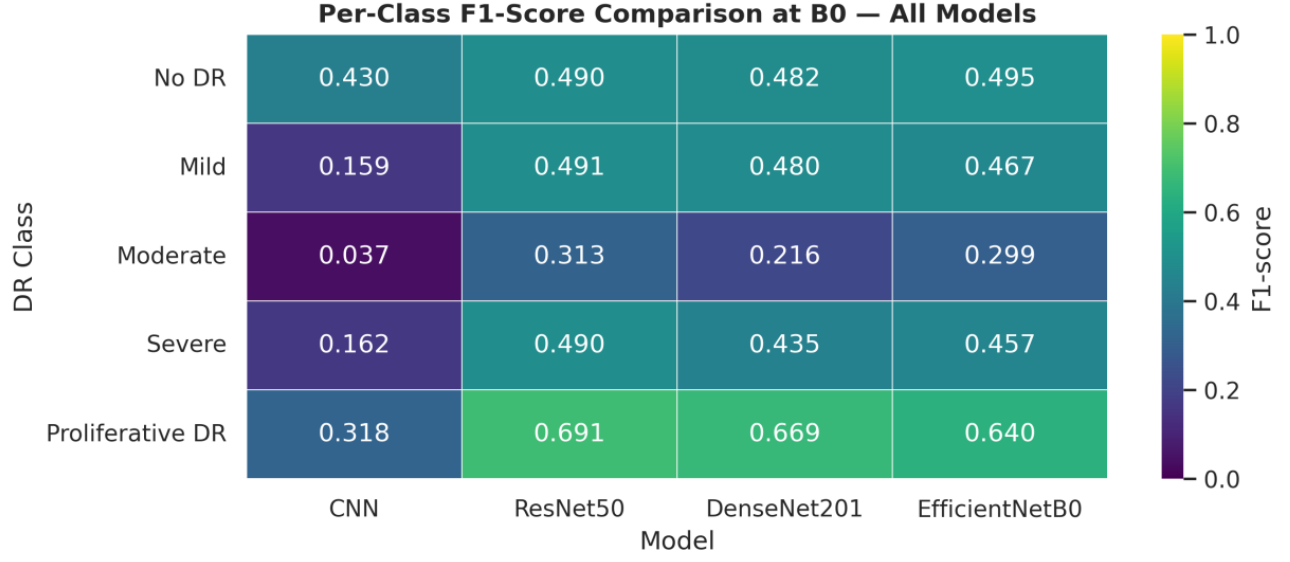


Fig. 8. Per-class F1 performance of all four models on the clean ($\sigma = 0$) test set.

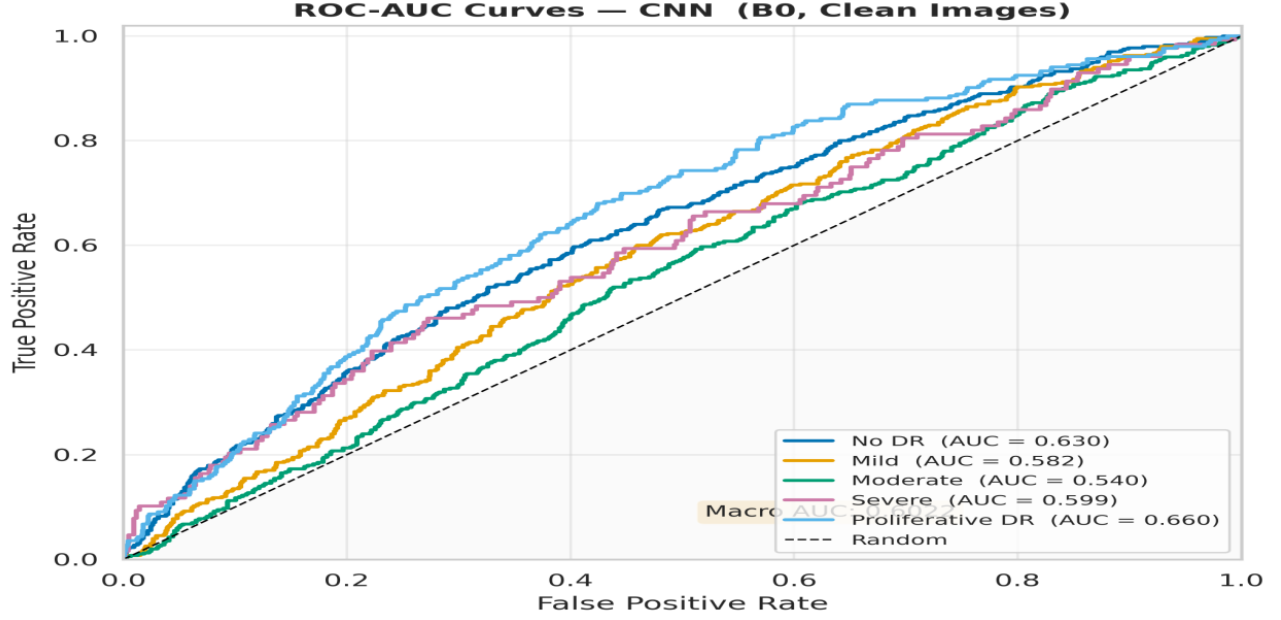


Fig. 9. One-vs-rest ROC curves for the pretrained models on clean ($\sigma = 0$) images.

TABLE IV
MACRO F1-SCORE FOR EACH MODEL ACROSS THE FIVE BLUR LEVELS.

Model	$\sigma = 0$	$\sigma = 1$	$\sigma = 2$	$\sigma = 3$	$\sigma = 4$
CNN	0.2211	0.2311	0.1790	0.1684	0.1656
ResNet50	0.4951	0.4260	0.2311	0.1090	0.0838
DenseNet201	0.4564	0.4495	0.2600	0.1745	0.1665
EfficientNetB0	0.4716	0.4400	0.3534	0.2471	0.2119

TABLE V
QUADRATIC WEIGHTED KAPPA FOR EACH MODEL ACROSS THE FIVE BLUR LEVELS.

Model	$\sigma = 0$	$\sigma = 1$	$\sigma = 2$	$\sigma = 3$	$\sigma = 4$
CNN	0.2076	0.2231	0.1477	0.1307	0.1187
ResNet50	0.6804	0.6030	0.2177	0.0365	0.0090
DenseNet201	0.6861	0.6203	0.2686	0.1382	0.0948
EfficientNetB0	0.6881	0.6287	0.4328	0.2676	0.1415

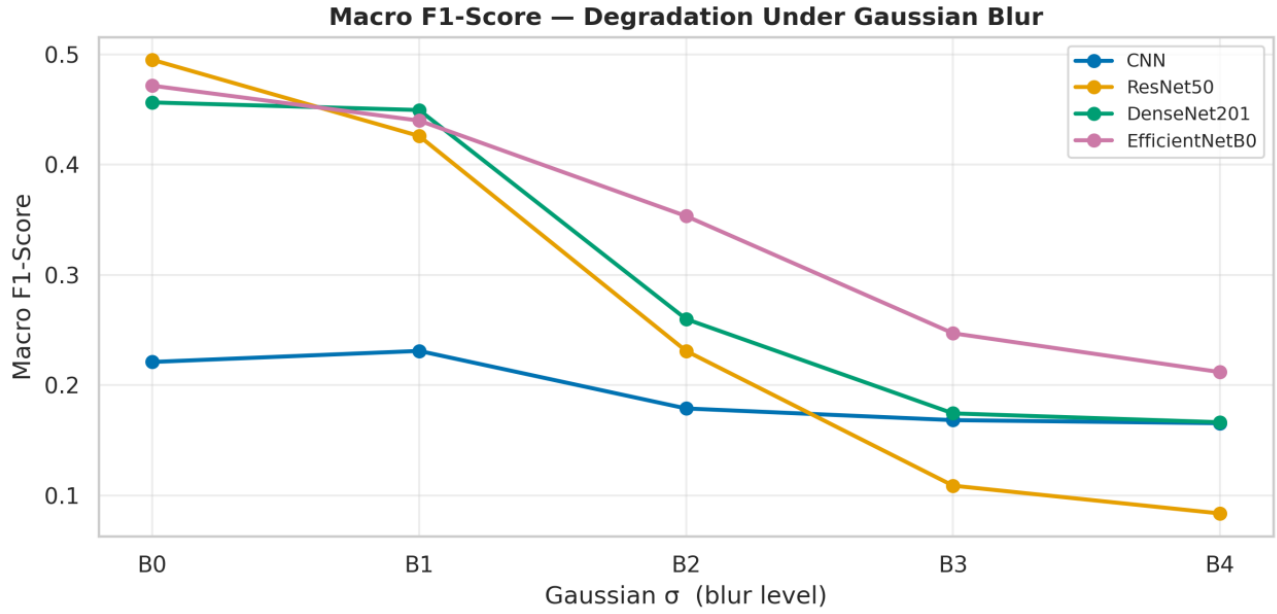


Fig. 10. Macro F1 degradation from $\sigma = 0$ to 4 for all four models.

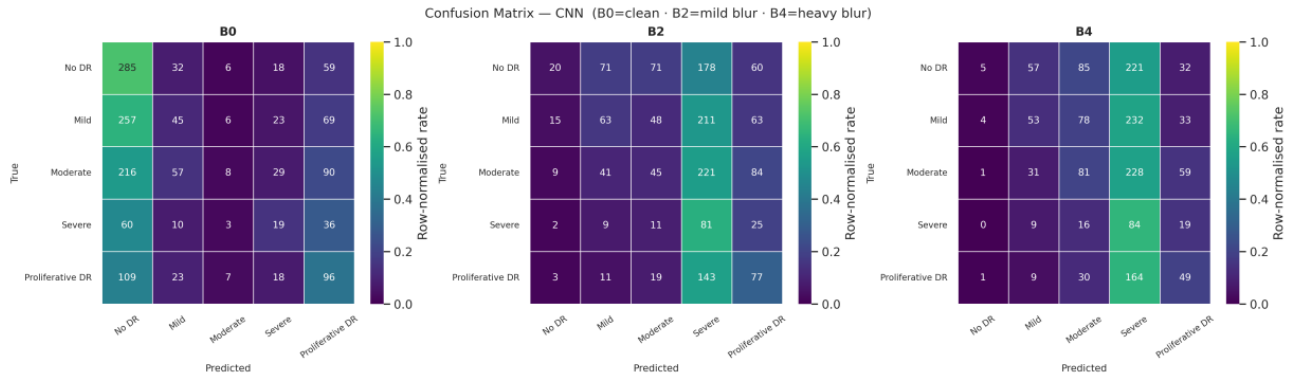


Fig. 11. Confusion matrices for CNN at $\sigma = 0$ (clean), $\sigma = 2$ (40% blur) and $\sigma = 4$ (80% blur). Cells show counts; shading shows the row-normalised rate.

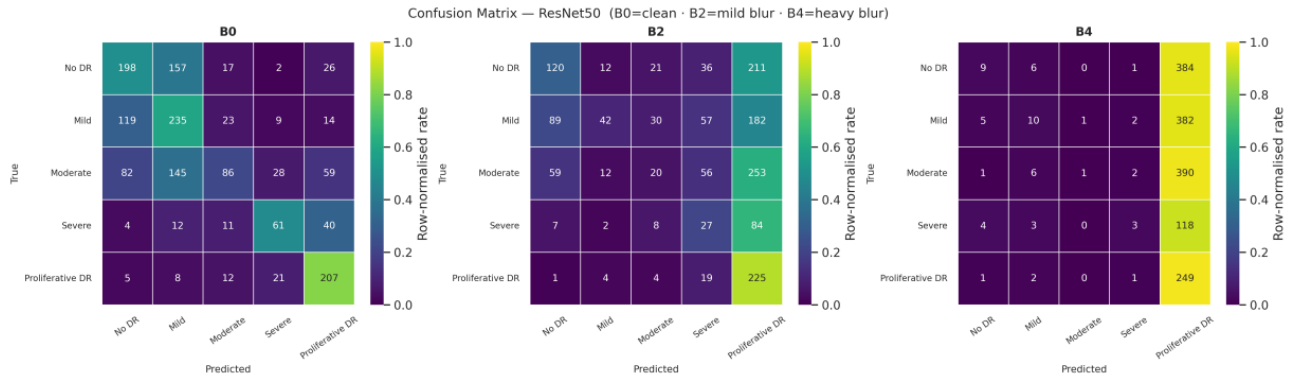


Fig. 12. Confusion matrices for ResNet50 at $\sigma = 0$ (clean), $\sigma = 2$ (40% blur) and $\sigma = 4$ (80% blur). Cells show counts; shading shows the row-normalised rate.

5) *Confusion Matrices Across Blur Levels:* Confusion matrices (Figures 11 to 14) were rendered at $\sigma = 0, 2$ and 4 only; those at $\sigma = 1$ and 3 were never generated and cannot be recovered. Error structure therefore cannot be described at the

two intermediate steps where Section 3.3.4 locates the largest single-step losses.

At $\sigma = 0$ the diagonal is strongest for No DR and Proliferative DR, weakest for Moderate DR, recalled at 0.21, 0.13 and 0.22

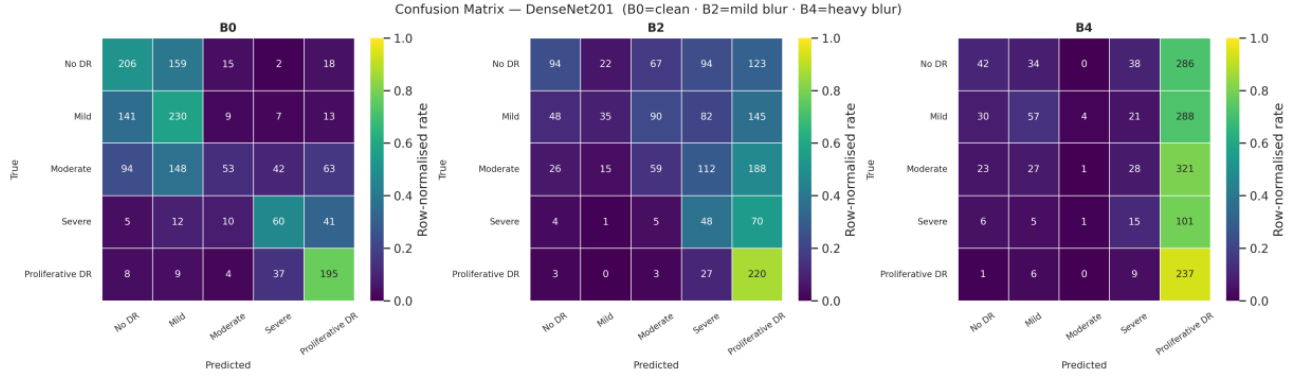


Fig. 13. Confusion matrices for DenseNet201 at $\sigma = 0$ (clean), $\sigma = 2$ (40% blur) and $\sigma = 4$ (80% blur). Cells show counts; shading shows the row-normalised rate.

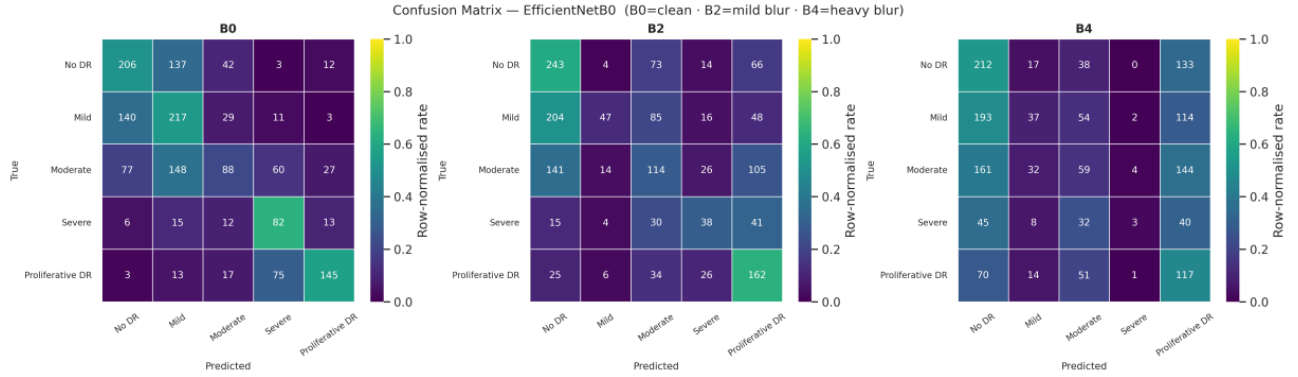


Fig. 14. Confusion matrices for EfficientNetB0 at $\sigma = 0$ (clean), $\sigma = 2$ (40% blur) and $\sigma = 4$ (80% blur). Cells show counts; shading shows the row-normalised rate.

by ResNet50, DenseNet201 and EfficientNetB0. The baseline CNN is already concentrated on one column: No DR recall 0.7125 against 0.1125 for Mild DR and 0.0200 for Moderate DR.

At $\sigma = 2$ and 4 prediction mass moves off the diagonal into higher-severity columns for three of the four models. At $\sigma = 4$ over-grading reaches 0.8134 for ResNet50, 0.7090 for DenseNet201 and 0.6603 for the CNN, with ResNet50's mean absolute grade error at 2.3011 grades. EfficientNetB0 is the single exception, and then only narrowly, under-grading 0.3839 of the test set against 0.3454 over-graded.

High Proliferative DR sensitivity at $\sigma = 4$ in the three over-grading models should not be read as class-level robustness. It is the signature of predictions collapsing onto the top class, occurring where ResNet50's QWK falls to 0.0090, the value expected of an effectively constant prediction.

6) *Exploratory Statistical Comparison:* Two sets of tests were performed on class-level metrics: whether the five per-class values recorded on clean images ($\sigma = 0$) differed between the four architectures (Table 6), and whether, within each model, those values differed across the five blur levels, $\sigma = 0$ to 4 (0% to 80% blur) (Table 7). Following supervision, the analysis was expanded from sensitivity alone to sensitivity, specificity, precision and F1-score, with a Tukey post hoc wherever the omnibus test was significant.

Sensitivity alone produced no significant difference between

architectures, $F(3, 16) = 1.29$, $p = .313$, which sat oddly beside ResNet50's QWK falling from 0.6804 to 0.0090. F1-score resolved the contradiction: blur significantly affected every pretrained model, ResNet50 $F(4, 20) = 11.01$, $p < .001$, DenseNet201 $F(4, 20) = 6.04$, $p = .002$ and EfficientNetB0 $F(4, 20) = 3.78$, $p = .019$, while the baseline CNN showed none, $F(4, 20) = 0.46$, $p = .762$. Sensitivity varies widely between severity classes and that variation dominates the error term, whereas F1-score, combining precision and recall, is less exposed to it. On clean images, precision differed significantly between architectures, $F(3, 16) = 8.04$, $p = .002$, with Tukey comparisons placing that difference entirely between the CNN and each pretrained model, and the three pretrained models statistically indistinguishable from one another. Each group comprises five class-level values from a single training run, so these are not independent replicates [34] and

the analysis is reported as exploratory rather than confirmatory. The CNN's null should also not be read as tolerance of blur: a model performing near chance has little to lose, and its stability reflects that rather than robustness.

7) *Grad-CAM Outputs:* Grad-CAM [28] was applied to all four architectures across $\sigma = 0$ to 4 (0% to 80% blur). Two cases were selected per model: an image classified correctly on clean input and an image misclassified on clean input. Holding each case constant across the ladder means only sharpness varies between panels. Attention change was quantified by the

TABLE VI
ONE-WAY ANOVA OF THE EFFECT OF MODEL ARCHITECTURE ON CLASS-LEVEL METRICS, CLEAN IMAGES ($\sigma = 0$). * $p < .05$.

Dependent variable	Source	Sum of Squares	df	Mean Square	F	Sig.
Sensitivity	Between	0.1976	3	0.0659	1.286	.313
	Within	0.8192	16	0.0512		
Specificity	Between	0.0092	3	0.0031	0.168	.917
	Within	0.2911	16	0.0182		
Precision	Between	0.2165	3	0.0722	8.038	.002*
	Within	0.1436	16	0.0090		
F1-score	Between	0.2444	3	0.0815	3.957	.028*
	Within	0.3293	16	0.0206		

TABLE VII
ONE-WAY ANOVA OF THE EFFECT OF BLUR LEVEL ON CLASS-LEVEL F1-SCORE, BY MODEL. * $p < .05$.

Model	Source	Sum of Squares	df	Mean Square	F	Sig.
CNN	Between Groups	0.0190	4	0.0048	0.463	.762
	Within Groups	0.2052	20	0.0103		
	Total	0.2242	24			
ResNet50	Between Groups	0.6857	4	0.1714	11.008	<.001*
	Within Groups	0.3114	20	0.0156		
	Total	0.9971	24			
DenseNet201	Between Groups	0.4100	4	0.1025	6.036	.002*
	Within Groups	0.3397	20	0.0170		
	Total	0.7497	24			
EfficientNetB0	Between Groups	0.2621	4	0.0655	3.779	.019*
	Within Groups	0.3467	20	0.0173		
	Total	0.6087	24			

overlap between the highest-activation region of each map and that of its own unblurred counterpart. The overlays are shown in Figure 15.

The overlay uses the viridis colour map: yellow marks the regions contributing most to the score of the predicted class, green an intermediate contribution, and purple and blue little or nothing. Activation values are rescaled to zero-to-one within each map before the colours are applied, so colour is a ranking within a single panel and carries no absolute meaning: no comparison of colour intensity is valid across panels, blur levels or architectures.

Attention overlap generally fell as blur increased for every model. For EfficientNetB0 on a correctly classified Moderate case, overlap fell from 1.000 to 0.130 across the ladder while the predicted grade stayed Moderate and confidence declined from 0.421 to 0.226, so prediction and attention can decouple. For the ResNet50 correct case the prediction moved from Mild at $\sigma = 0$ to Moderate, then to Proliferative DR at $\sigma = 3$ and 4, while confidence rose from 0.571 to 0.968; the model became more certain as it became progressively more wrong, which is the behaviour an operator would find hardest to detect in deployment. The CNN's correct case shows the sharpest loss, its overlap falling from 1.000 to 0.000 as the prediction moves from Mild to Severe. These findings rest on two cases per architecture and are reported as illustrative rather than as a general characterisation of attention behaviour.

The complete Grad-CAM outputs, with per-panel predicted grade, confidence and attention- overlap values, were retained with the project artefacts; interpretation follows in Chapter 4.

IV. DISCUSSION OF FINDINGS

A. Findings

RQ1: How does progressively increasing Gaussian blur affect five-class DR grading performance on the same held-out images?

Macro F1-score, quadratic weighted kappa (QWK) and macro ROC-AUC all fell between $\sigma = 0$ and 4 (80% blur) for all four architectures, the pretrained models losing most from the highest clean-image starting points. EfficientNetB0 fell from 0.4716 to 0.2119; ResNet50 from the strongest clean-image macro F1 of 0.4951 to 0.0838; DenseNet201 from 0.4564 to 0.1665. The hypothesis was supported.

Blur level significantly affected F1-score for every pretrained model: ResNet50 $F(4, 20) = 11.01$, $p < .001$; DenseNet201 $F(4, 20) = 6.04$, $p = .002$; EfficientNetB0 $F(4, 20) = 3.78$, $p = .019$. The baseline CNN showed no significant effect, $F(4, 20) = 0.46$, $p = .762$, consistent with a model with little discriminative performance to lose. The earlier sensitivity-only analysis missed this because sensitivity varies widely between severity classes, and that variation dominates the error term.

RQ2: Which severity classes are most sensitive to blur, and is the error direction under-grading, over-grading, or adjacent confusion?

Degradation was not uniform across severity classes. Moderate DR was consistently among the weakest across the pretrained models, its sensitivity low on clean images and lower still under blur. Intermediate grades appear to depend on smaller, less visually dominant lesions that blur destroys first, leaving neighbouring grades hard to separate.

High sensitivity for proliferative DR under blur does not indicate class robustness: a model assigning almost every image to the most severe grade records near-complete sensitivity while misgrading the rest. At $\sigma = 4$ ResNet50 over-graded at 0.8134 while its QWK fell to 0.0090, ordinal agreement indistinguishable from chance. Per-class values, available at every blur level, show proliferative sensitivity rising as agreement with the reference grading collapses.

On clean images every architecture erred towards lower severity, but this reverses as sharpness is removed: at $\sigma = 4$ the CNN, ResNet50 and DenseNet201 over-graded 0.6603,

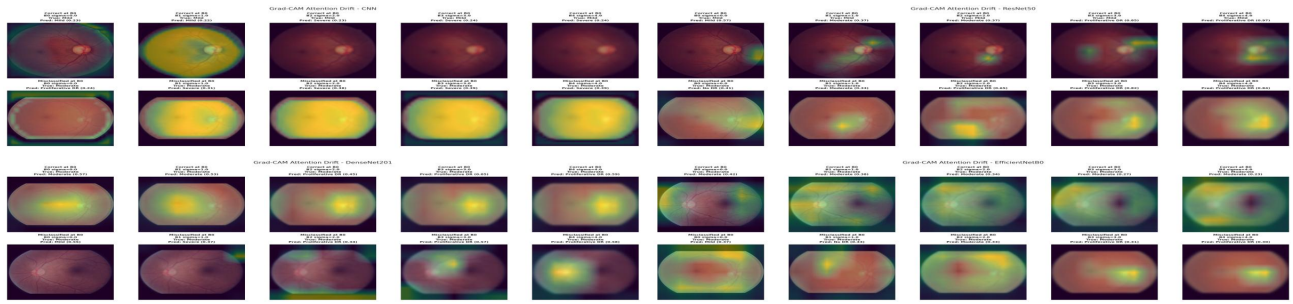


Fig. 15. Grad-CAM overlays across $\sigma = 0$ to 4 (0% to 80% blur) for a correctly classified and a misclassified case. The viridis map is used: yellow marks the strongest contribution to the predicted class, green an intermediate contribution, and purple and blue little or none. Activations are normalised within each panel, so colours are not comparable between panels. Predicted grade and confidence are printed above each panel.

0.8134 and 0.7090 of the test set, while only EfficientNetB0 still erred towards lower severity, and then only narrowly, at 0.3839 against 0.3454 over-grading. The dominant failure mode is not missed disease but false referral and over-treatment.

RQ3: Do the baseline CNN, ResNet50, DenseNet201 and EfficientNetB0 differ in their robustness to blur?

The baseline CNN reached a clean-image macro F1-score of only 0.2211, against 0.4564 to 0.4951 for the pretrained models, among which ResNet50 achieved the highest macro F1 and EfficientNetB0 the highest QWK. That ordering reverses under blur: EfficientNetB0 held its performance best, ResNet50 fell furthest.

Relative retention, given in Table 8, is each model’s macro F1 at $\sigma = 4$ as a fraction of its $\sigma = 0$ value; approximately 0.193 is chance level for this test set’s class support. The CNN’s retention is uninterpretable: its clean-image value sits only marginally above that floor and its $\sigma = 4$ value of 0.1656 falls below it, so the ratio describes movement within noise, not retained discriminative skill. The comparison is also unequal: the CNN received ten head epochs and no fine-tuning, each pretrained model ten head epochs plus up to twenty fine-tuning epochs. Retention should therefore be read only across the three pretrained architectures, among which EfficientNetB0 preserved the largest fraction of its clean-image performance.

RQ4: How does increasing blur alter the location and stability of Grad-CAM attention?

Grad-CAM was used on one correctly classified image and one incorrectly classified image for each network type throughout the blur ladder, with attention shift calculated as the overlap with its unblurred version. The overlap reduced for each of the networks studied as sharpness was reduced: the effect of degradation is not just changing the output but changing what the network focuses on. For EfficientNetB0, the grade was correct in all five versions while overlap decreased from 1.000 to 0.130: attention can migrate without the label changing. For ResNet50 the prediction drifted from Mild to Proliferative DR while confidence rose from 0.571 to 0.968, the model growing more certain as it grew less correct. With two cases per architecture the evidence is illustrative rather than general, but measured rather than asserted.

Model $\sigma = 4$ F1 Retention

CNN 0.749

EfficientNetB0 0.449

DenseNet201 0.365

ResNet50 0.169

Image quality degradation is therefore material to automated DR grading, and clean-benchmark evaluation understates deployed error rates.

B. Evaluation of the Product

The product delivered by this project is an experimental pipeline dataset sampling, preprocessing, deterministic blur generation, training and evaluation together with the four classifiers it produced. It is assessed here against the purpose it was built for, which was to measure how a classifier trained on clean fundus images behaves when image quality falls at deployment, rather than to produce a deployable grading system.

The pipeline has four defensible strengths. First, one protocol was applied to all four architectures: the same padded 224×224 inputs, the same sampled splits of 7,006 training, 1,574 validation and 1,581 test images, the same batch size, optimiser, loss and callbacks, and the same five-point test-time blur ladder. Differences between models can therefore be attributed to architecture and to training budget rather than to inconsistency in the evaluation. Second, the pipeline was deterministically seeded, with a fixed seed of 42 applied to Python, NumPy, TensorFlow and dataset sampling, so that the subset, the shuffle order and the generated blur can be reproduced from the submitted notebook. Third, the clean-train / degraded-test design isolates the quantity of interest. Blur was applied only when the test copies

were created and never during training or validation, so what was measured is the deployment- time degradation of a fixed classifier and not the training-time robustness that blur augmentation would confer. These are separate questions, and the design does not confound them. Fourth, the blur ladder was verified by two metrics of different type rather than by visual inspection alone. Median Laplacian variance, a no-reference measure of edge strength within a single image, fell monotonically from 342.12 at $\sigma = 0$ to 2.56 at $\sigma = 4$ (80% blur), while mean PSNR, a full-reference measure against the original, fell from 34.086 dB at $\sigma = 1$ to 27.514 dB at $\sigma = 4$. Agreement between an internal and a referenced measure is stronger evidence than either alone.

Five weaknesses are equally clear. The σ values were calculated for images of about 300 pixels but used on resized

images of 224×224 , which means that the degradation introduced is more severe compared to the calibration intended. The bracketed percentages indicate the steps on the σ ladder and do not represent the calibrated severities. No grid search or random search has been conducted, but only a few settings have been determined via pilot comparison, and each of the backbones was trained on one set of the settings, so the results of comparison include the benefits that such settings offer to one backbone over another. Sharpness statistics have been calculated for the entire padded image and thus incorporate both retinal details and artificial edge along with a lot of empty space, so that the values cannot be considered estimates of retinal sharpness. The evaluation subset destroys the natural class prior, with the consequences set out in Section 4.4. Finally, the baseline CNN is not effort-matched: it was trained for 10 epochs with no fine-tuning phase, against 10 head epochs plus up to 20 fine-tuning epochs for the three pretrained models, so it functions as a lower reference point rather than as a controlled architectural comparator.

Several tools and methods were available but were not adopted. TensorFlow and Keras were used throughout [20] in preference to PyTorch, because the Keras Applications interface supplies the matched preprocessing function for each pretrained backbone and a single framework removed one source of inconsistency. Grad-CAM was preferred to perturbation-based explainers such as LIME or SHAP because it requires only a forward and backward pass and has established use in DR classification [28, 16], whereas perturbation methods would have added substantial computation across five blur levels and four models. Per-class undersampling was used in preference to oversampling or synthetic minority generation, which would have repeated or fabricated images in the classes where there is the least amount of variation. The restoration front end was not chosen due to it testing deblurring instead of the sensitivity of the classifier, and transformers were never tested due to the insufficient amount of training data and free GPU resources.

C. Evaluation of the Project Process

This section will assess the way the project was carried out as opposed to the findings of the project itself. First person will be utilized in this section since the requirements of the assessment ask for a reflection on my decisions.

Achievement of the Terms of Reference objectives

Assessment follows the ten Terms of Reference (ToR) objectives, not their narrower restatement as eight in section 1.3. Objectives 1, 3, 5, 8, 9 and 10 (literature review, model training, Grad-CAM visualisations, comparative analysis, documentation) were met, two with qualification. The baseline CNN ran ten head epochs with no fine-tuning phase, against ten head plus up to twenty fine-tuning epochs per pretrained model, so it was never held to the budget of the models it benchmarks. The Grad-CAM analysis covers two cases per architecture rather than a stratified sample, so it quantifies attention change without characterising it across the dataset. Objective 4 was met only in part: the published labels could not be independently verified.

Objective 2, to define the functional and experimental requirements for evaluating multi-class DR classification models,

was not carried out and appears nowhere in this document; requirements were settled informally as the pipeline was written. This is the clearest failure of process in the project: fixed at the start, they would have settled the sampling budget, the blur ladder and the unit of analysis before any model was trained.

Objective 6 required accuracy, precision, recall, F1-score and confusion matrices; while the former four metrics were provided in their entirety, the latter two could not be, since the plotting function generates plots only for $\sigma = 0, 2$ and 4 , meaning that the matrices for $\sigma = 1$ (20% blur) and $\sigma = 3$ (60% blur) could not have been created.

Objective 7 called for blur at 0%, 20% and 40%. While I over-performed on the ladder, creating five deterministic steps from $\sigma = 0$ (0% blur) to $\sigma = 4$ (80% blur), which would otherwise have hidden the non-linear nature of the relationship, I under-performed on the coverage front, limiting degradation to just the test set.

Plan against actual progress

The project broadly followed the ToR plan, though the experimental scope grew beyond it: three blur levels were originally specified, while five were carried out across four models. An approved extension provided additional time, which was used for analysis and writing

Resources, contingency and a supervised design change

The ToR named Google Colab as primary, with the fallback "Kaggle Notebooks or university computing facilities if usage limits are reached". That fallback was needed and worked: when the Colab allowance became limiting, the pipeline moved to Kaggle Notebooks without redesign. It was not free: Kaggle could not support repeated training over the 73,995-image training pool either, so FAST_MODE was enabled and a capped per-class subset drawn, retaining 7,006 training images and discarding 90.5% of the pool and 97.3% of the No DR class. Every absolute figure in Chapter 3 carries that cost, and the consequence shows in section 3.3.6, where the sensitivity-only analysis detected nothing.

The ladder was originally to reach 100% blur; after discussion with my supervisor it was capped at $\sigma = 4$ (80% blur). Median Laplacian variance had already fallen from 342.12 at $\sigma = 0$ to 2.56 at $\sigma = 4$, so a further level would have added almost nothing discriminable. I record this as a supervised design change, not a silent deviation.

Legal, social, ethical and professional issues

The ethics approval statement for this project claimed no human subjects and publicly available secondary data, the single, merged EyePACS, APTOS and Messidor dataset on Kaggle [1], without handling any identifiable information at any time. This was true, but the process of fulfilling it revealed the truth behind the statement: anonymized is not independent. There is no table of patients in the dataset, so I made up a proxy patient ID by analyzing image file names and compared overlap in these IDs across splits; the numbers reported in 3.2.2 were positive for all three pairs. I could not verify it: the ID was a guess, not a checkable fact, and there was nothing in the metadata that would help me with verification or rejection. I included it as a limitation and did not use the term "patient independent" regarding any splits in this report. An integrity issue that cannot be verified in a medical dataset is an issue

itself; hiding it would turn all the performance numbers into overclaims.

What I would do differently

First, I would establish the sampling budget prior to training; decided upon reactively after the compute limit was reached, it ended up throwing away 90.5% of the pool automatically rather than classifying one class in trade for another based on blur level. Second, I would determine the unit of analysis in advance; the experiments in section 3.3.6 were performed using five per-class values of each model, and my choice of sensitivity as the only dependent variable obscured an effect detectable via the F1-score. Third, I would compute sharpness on a retinal mask, since the circle representing the field of view and black background contribute Laplacian variance but carry no retinal information. Fourth, I would budget for multiple random seeds, as the single fixed seed leaves every difference between models confounded with initialisation.

What these four shares is timing rather than difficulty: each was settled late and under pressure when it could have been settled cheaply at the start, and that ordering limited what the evidence could support.

D. Limitations

The evaluation subset does not carry the class prior of a screening population. No DR accounts for 21.4% of the sampled training images and 400 of the 1,581 test images, whereas 55,162 of the 73,995 images in the source training pool, or 74.5%, are No DR. Every precision-based quantity reported here is therefore conditioned on an artificial base rate, and no precision or positive predictive value from these results transfers to a screening setting, where the same sensitivity would return far lower precision. Recall-based measures and QWK are less exposed to this, but no claim about screening yield can rest on the figures presented.

The subset was obtained by discarding 90.5% of the available training images, 7,006 being retained from 73,995, and 97.3% of the No DR class. Class imbalance was consequently handled by that sampling rather than by the weighting described in Section 3.2.4; the weights actually applied were 0.934 for four classes and 1.393 for Severe, which is close to inert.

The statistical design remains a limitation. Each group comprises five class-level values from a single training run, treated as independent observations, so the tests describe variation between classes rather than between runs and are exploratory rather than confirmatory even where they reach significance. The original analysis also used sensitivity as its only dependent variable, which is the noisiest class-level metric across severity grades; expanding to F1-score detected a blur effect that the sensitivity-only analysis had missed entirely.

No referable-DR analysis at grade ≥ 2 was computed at any point in this project, although referable disease is the decision a screening programme actually makes. The five-class framing therefore leaves the clinically operative question unaddressed.

Moderate DR recall lies between 0.02 and 0.22 at $\sigma = 0$, before any blur is applied: 0.02 for the CNN, 0.13 for DenseNet201, 0.21 for ResNet50 and 0.22 for EfficientNetB0. The blur experiment consequently measures the degradation

of classifiers that were already failing on the class separating referable from non-referable disease, and the Moderate-class findings should be read with that in mind.

Two constraints noted above also limit interpretation: the σ ladder was calibrated for approximately 300-pixel images but applied at 224 x 224, and the Grad-CAM analysis covers two cases per architecture, which is enough to quantify attention change but not to characterise it across the dataset.

V. CONCLUSIONS AND RECOMMENDATIONS

This project quantified how far controlled loss of fundus image quality alters automated diabetic retinopathy (DR) grading. A custom CNN baseline, ResNet50, DenseNet201 and EfficientNetB0 were trained on clean images and tested on one held-out set re-rendered at five Gaussian blur levels, $\sigma = 0$ to 4 (80% blur), with image quality the only variable.

Blur reduced performance for every architecture (RQ1): averaged over the three pretrained models, macro F1 fell 67.5% and QWK 88.1%, non-linearly, with the knee between $\sigma = 1$ and 3.

Error direction also changed (RQ2): at $\sigma = 4$ three of four models over-graded, ResNet50 most heavily at 81.3% of test images; EfficientNetB0 alone still erred towards lower severity, and then only narrowly. Apparent proliferative DR robustness reflects collapse onto the most severe grade, confirmed by ResNet50's QWK of 0.0090.

Robustness differed by architecture (RQ3): EfficientNetB0 retained the largest share of clean macro F1 at $\sigma = 4$, ahead of DenseNet201 and ResNet50. The CNN's higher apparent retention

is an artefact: its macro F1 there lies below the approximate chance floor of 0.193, and it received one training phase to the pretrained models' two.

Attention moved as blur increased (RQ4): Grad-CAM overlap with the unblurred map fell for every architecture, and for ResNet50 confidence rose as its prediction drifted from the reference grade. With two cases per architecture the evidence is illustrative rather than general, but quantified rather than described.

Synthetic Gaussian blur does not reproduce the optical defocus, motion and illumination faults of real portable imaging. All images came from one aggregated source, no external validation was performed, and training used 7,006 of 73,995 images. These tests contrast five class values per condition per single run and are exploratory in nature, even in cases where significance is found. The hyperparameters used were determined from a pilot study and do not come from an exhaustive search and the referable-DR performance metric has not been calculated.

There should be a quality control gate before the image grading process: the performance remains relatively consistent up to a point of $\sigma = 1$ (median Laplacian variance of 21.48, mean PSNR of 34.086 dB), and degrades more rapidly between $\sigma = 1$ and 3 (3.21, 28.731 dB). Images with blurring beyond $\sigma = 1$, or those with a Laplacian variance lower than 21.48, should be rejected for grading and re-captured, as the Laplacian metric covers the entire image. Suggested extensions include: training

for blurred images, testing on realistically degraded images and smartphones, external validation, and expert comparison of lesion markings vs Grad-CAM maps.

Image quality assessment does not determine performance on degraded images; robustness should be tested on degraded images for each deployed network and reliability claims reported with the quality range over which they hold.

ACKNOWLEDGMENT

I would like to express my sincere gratitude to everyone who contributed to the successful completion of this project. First, I would like to thank my supervisor for their continuous guidance, valuable feedback, and support throughout the development of this work. Their advice and encouragement greatly helped me improve the quality of this project. I would also like to acknowledge the support and assistance provided by the staff members of the department, whose knowledge and guidance contributed to the successful completion of this project.

REFERENCES

- [1] Ascanipek, A.S. (2024) EyePACS, APTOS, Messidor Diabetic Retinopathy [Dataset]. Kaggle. Available at: <https://www.kaggle.com/datasets/ascanipek/eyepacs-aptos-messidor-diabetic-retinopathy> (Accessed: 23 August 2026).
- [2] World Health Organization [2] Diabetes. Fact sheet, 14 November. Geneva: World Health Organization. Available at: <https://www.who.int/news-room/fact-sheets/detail/diabetes> (Accessed: 31 August 2026).
- [3] NHS England [3] Diabetic eye screening. Available at: <https://www.nhs.uk/tests-and-treatments/diabetic-eye-screening/> (Accessed: 31 August 2026).
- [4] Qureshi, I., Ma, J. and Abbas, Q. (2021) 'Diabetic retinopathy detection and stage classification in eye fundus images using active deep learning', *Multimedia Tools and Applications*, 80(8), pp. 11691–11721. doi:10.1007/s11042-020-10238-4.
- [5] Mehboob, A., Akram, M.U., Alghamdi, N.S. and Abdul Salam, A. (2022) 'A deep learning based approach for grading of diabetic retinopathy using large fundus image dataset', *Diagnostics*, 12(12), 3084. doi:10.3390/diagnostics12123084.
- [6] Vaish, A., Singh, D., Garg, A., Tiwari, A. and Rathore, A. (2024) 'Classification of diabetic retinopathy using improved ResNet-50 deep learning model', 2024 Third International Conference on Distributed Computing and Electrical Circuits and Electronics (ICDCECE). Ballari, India, 26–27 April. Piscataway, NJ: IEEE, pp. 1–5. doi:10.1109/ICDCECE60827.2024.10549184.
- [7] Moodu, V., Prasad, C.R. and Rebelli, S. (2025) 'Optimized deep learning architectures using EfficientNet for diabetic retinopathy diagnosis', 2025 International Conference on World Summit on Unmanned Aerial Systems (WorldSUAS). Indore, India. Piscataway, NJ: IEEE, pp. 1–5. doi:10.1109/WorldSUAS66815.2025.11199165.
- [8] Liu, H., Li, H., Wang, X., Li, H., Ou, M., Hao, L., Hu, Y. and Liu, J. (2022) 'Understanding how fundus image quality degradation affects CNN-based diagnosis', 2022 44th Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC). Glasgow, UK, 11–15 July. Piscataway, NJ: IEEE, pp. 438–442. doi:10.1109/EMBC48229.2022.9871507.
- [9] Iqbal, U. (2021) 'Smartphone fundus photography: a narrative review', *International Journal of Retina and Vitreous*, 7, 44. doi:10.1186/s40942-021-00313-9.
- [10] Chalam, K.V., Chamchikh, J. and Gasparian, S. (2022) 'Optics and utility of low-cost smartphone-based portable digital fundus camera system for screening of retinal diseases', *Diagnostics*, 12(6), 1499. doi:10.3390/diagnostics12061499.
- [11] Wilkinson, C.P., Ferris, F.L., Klein, R.E., Lee, P.P., Agardh, C.D., Davis, M., Dills, D., Kampik, A., Pararajasegaram, R. and Verdager, J.T. (2003) 'Proposed international clinical diabetic retinopathy and diabetic macular edema disease severity scales', *Ophthalmology*, 110(9), pp. 1677–1682. doi:10.1016/S0161-6420(03)00475-5.
- [12] Shi, D., Zhang, W., He, S., Chen, Y., Song, F., Liu, S., Wang, R., Zheng, Y. and He, M. (2023) 'Translation of color fundus photography into fluorescein angiography using deep learning for enhanced diabetic retinopathy screening', *Ophthalmology Science*, 3(4), 100401. doi:10.1016/j.xops.2023.100401.
- [13] Reddiar Seetharaman, K.M. (2025) 'Automated eye disease detection of diabetic retinopathy using artificial intelligence on fundus images', 2025 International Conference on Networks and Cryptology (NETCRYPT). New Delhi, India, 29–31 May. Piscataway, NJ: IEEE, pp. 590–595. doi:10.1109/NETCRYPT65877.2025.11102467.
- [14] Matthew, A., Gunawan, A.A.S. and Kurniadi, F.I. (2023) 'Diabetic retinopathy diagnosis system based on retinal biomarkers using EfficientNet-B0 for Android devices', 2023 IEEE International Conference on Communication, Networks and Satellite (COMNETSAT). Malang, Indonesia, 23–25 November. Piscataway, NJ: IEEE, pp. 207–212. doi:10.1109/COMNETSAT59769.2023.10420736.
- [15] Sornil, A.B., Sheeja Herobin Rani, C. and Sheeba, I.R. (2024) 'Predicting diabetic retinopathy severity with deep learning: a survey of fundus image analysis technique', 2024 10th International Conference on Communication and Signal Processing (ICCS). Melmaruvathur, India, 12–14 April. Piscataway, NJ: IEEE, pp. 1274–1278. doi:10.1109/ICCS60870.2024.10543945.
- [16] Muntaha, S., Salam, S.S. and Mustakim, N. (2024) 'An explainable AI-based deep learning model for classification of diabetic retinopathy stages using retinal fundus images', 2024 27th International Conference on Computer and Information Technology (ICCIT). Cox's Bazar, Bangladesh, 20–22 December. Piscataway, NJ: IEEE, pp. 3010–3015. doi:10.1109/ICCIT64611.2024.11022403.
- [17] Hendrycks, D. and Dietterich, T.G. (2019) 'Benchmarking neural network robustness to common corruptions and perturbations', 7th International Conference on Learning Representations (ICLR 2019). New Orleans, LA, 6–9 May. Available at: <https://arxiv.org/abs/1903.12261>.
- [18] Dodge, S. and Karam, L. (2016) 'Understanding how image quality affects deep neural networks', 2016 Eighth International Conference on Quality of Multimedia Experience (QoMEX). Lisbon, Portugal, 6–8 June. Piscataway, NJ: IEEE, pp. 1–6. doi:10.1109/QoMEX.2016.7498955.
- [19] Vasiljevic, I., Chakrabarti, A. and Shakhnarovich, G. (2016) Examining the impact of blur on recognition by convolutional networks. arXiv:1611.05760. Available at: <https://arxiv.org/abs/1611.05760>.
- [20] Abadi, M., Barham, P., Chen, J., Chen, Z., Davis, A., Dean, J., Devin, M., Ghemawat, S., Irving, G., Isard, M., Kudlur, M., Levenberg, J., Monga, R., Moore, S., Murray, D.G., Steiner, B., Tucker, P., Vasudevan, V., Warden, P., Wicke, M., Yu, Y. and Zheng, X. (2016) 'TensorFlow: a system for large-scale machine learning', *Proceedings of the 12th USENIX Symposium on Operating Systems Design and Implementation (OSDI '16)*. Savannah, GA, 2–4 November. Berkeley, CA: USENIX Association, pp. 265–283.
- [21] Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M. and Duchesnay, É. (2011) 'Scikit-learn: machine learning in Python', *Journal of Machine Learning Research*, 12, pp. 2825–2830.
- [22] Cuadros, J. and Bresnick, G. (2009) 'EyePACS: an adaptable telemedicine system for diabetic retinopathy screening', *Journal of Diabetes Science and Technology*, 3(3), pp. 509–516. doi:10.1177/193229680900300315.
- [23] Decencière, E., Zhang, X., Cazuguel, G., Laÿ, B., Cochenier, B., Trone, C., Gain, P., Ordóñez, R., Massin, P., Erginay, A., Charton, B. and Klein, J.-C. (2014) 'Feedback on a publicly distributed image database: the Messidor database', *Image Analysis & Stereology*, 33(3), pp. 231–234. doi:10.5566/ias.1155.
- [24] He, K., Zhang, X., Ren, S. and Sun, J. (2016) 'Deep residual learning for image recognition', 2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR). Las Vegas, NV, 27–30 June. Piscataway, NJ: IEEE, pp. 770–778. doi:10.1109/CVPR.2016.90.
- [25] Huang, G., Liu, Z., Van der Maaten, L. and Weinberger, K.Q. (2017) 'Densely connected convolutional networks', 2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR). Honolulu, HI, 21–26 July. Piscataway, NJ: IEEE, pp. 2261–2269. doi:10.1109/CVPR.2017.243.
- [26] Tan, M. and Le, Q.V. (2019) 'EfficientNet: rethinking model scaling for convolutional neural networks', *Proceedings of the 36th International Conference on Machine Learning, PMLR*, 97, pp. 6105–6114.
- [27] Deng, J., Dong, W., Socher, R., Li, L.-J., Li, K. and Fei-Fei, L. (2009) 'ImageNet: a large-scale hierarchical image database', 2009 IEEE Conference on Computer Vision and Pattern Recognition (CVPR). Miami, FL, 20–25 June. Piscataway, NJ: IEEE, pp. 248–255. doi:10.1109/CVPR.2009.5206848.

- [28] Selvaraju, R.R., Cogswell, M., Das, A., Vedantam, R., Parikh, D. and Batra, D. (2017) ‘Grad- CAM: visual explanations from deep networks via gradient-based localization’, 2017 IEEE International Conference on Computer Vision (ICCV). Venice, Italy, 22–29 October. Piscataway, NJ: IEEE, pp. 618–626. doi:10.1109/ICCV.2017.74.
- [29] Pech-Pacheco, J.L., Cristóbal, G., Chamorro-Martínez, J. and Fernández-Valdivia, J. (2000) ‘Diatom autofocusing in brightfield microscopy: a comparative study’, Proceedings of the 15th International Conference on Pattern Recognition (ICPR 2000). Barcelona, Spain, 3–7 September. Piscataway, NJ: IEEE, vol. 3, pp. 3318–3321. doi:10.1109/ICPR.2000.903548.
- [30] Horé, A. and Ziou, D. (2010) ‘Image quality metrics: PSNR vs. SSIM’, 2010 20th International Conference on Pattern Recognition (ICPR). Istanbul, Turkey, 23–26 August. Piscataway, NJ: IEEE, pp. 2366–2369. doi:10.1109/ICPR.2010.579.
- [31] Kingma, D.P. and Ba, J. (2015) ‘Adam: a method for stochastic optimization’, 3rd International Conference on Learning Representations (ICLR 2015). San Diego, CA, 7–9 May. Available at: <https://arxiv.org/abs/1412.6980>.
- [32] Yosinski, J., Clune, J., Bengio, Y. and Lipson, H. (2014) ‘How transferable are features in deep neural networks?’, Advances in Neural Information Processing Systems 27 (NIPS 2014). Montreal, Canada, 8–13 December, pp. 3320–3328.
- [33] Ioffe, S. and Szegedy, C. (2015) ‘Batch normalization: accelerating deep network training by reducing internal covariate shift’, Proceedings of the 32nd International Conference on Machine Learning, PMLR, 37, pp. 448–456.
- [34] Hurlbert, S.H. (1984) ‘Pseudoreplication and the design of ecological field experiments’, *Ecological Monographs*, 54(2), pp. 187–211. doi:10.2307/1942661.